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Predicting multiple conformations via sequence clustering and AlphaFold2

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Abstract. AlphaFold2 (AF2)¹ has revolutionized structural biology by accurately predicting single structures of proteins. However, a protein's biological function often depends on multiple conformational substates², and disease-causing point mutations often cause population changes within these substates^{3,4}. We demonstrate that clustering a multiple sequence alignment (MSA) by sequence similarity enables AF2 to sample alternate states of known metamorphic proteins with high confidence. Using this method, AF-Cluster, we investigated the evolutionary distribution of predicted structures for the metamorphic protein KaiB⁵, and found that predictions of both conformations were distributed in clusters across the KaiB-family. We used nuclear magnetic resonance (NMR) spectroscopy to confirm a surprising AF-Cluster prediction: a cyanobacteria KaiB variant is stabilized in the opposite state than the more widely-studied variant. To test AF-Cluster's sensitivity to point mutations, we designed and experimentally verified a set of 3 mutations predicted to flip KaiB from *Rhodobacter sphaeroides* from the ground to the fold-switched state. Finally, screening for alternate states in protein families without known fold-switching identified a putative alternate state for the oxidoreductase Mpt53 in *M. tuberculosis*. Further development of such bioinformatic methods in tandem with experiments will likely have profound impact on predicting protein energy landscapes, essential for illuminating biological function.

Introduction

Understanding the mechanistic basis of any protein's functions requires understanding the complete set of conformational substates it can adopt². For any protein structure prediction method, the task of predicting ensembles can be considered in two parts: an ideal method would (1) generate conformations encompassing the complete landscape and (2) score these conformations in accordance with the underlying Boltzmann distribution. AlphaFold2 (AF2) achieved breakthrough performance in the CASP14 competition⁶ in part by advancing the state of the art for inferring patterns of interactions between related sequences in a MSA, building on a long history of methods for inferring these patterns⁷⁻¹⁰, often called evolutionary couplings. The premise of methods to infer structure based on evolutionary couplings is that because amino acids exist and evolve in the context of 3D structure, they are not free to evolve independently, but instead co-evolve in patterns reflective of the underlying structure. However, proteins must evolve in the context of the multiple conformational states they adopt. AF2's unprecedented accuracy¹ at single-structure prediction has garnered interest in its ability to predict multiple conformations of proteins, yet AF2 has been demonstrated to fail in predicting multiple structures of metamorphic proteins¹¹, proteins with apo/holo conformational changes¹², and other multi-state proteins¹³ using its default settings. Despite these demonstrations of shortcomings, Del Alamo et al. have shown that subsampling the input MSA allows AF2 to predict known conformational changes of an ion channel¹⁴.

Why does subsampling an MSA lead to predicting multiple conformational states? Success of this approach in a given system implies that when calculating evolutionary couplings with a complete MSA, evolutionary couplings for multiple states are already sufficiently present such that when the MSA is subsampled, introducing noise that will obscure subsets of these contacts, there are still sufficiently complete sets of contacts corresponding to one or the other state. Indeed, methods for inferring evolutionary couplings have already demonstrated that contacts corresponding to multiple states can be observed at the level of entire MSAs for membrane proteins¹⁵, ligand-induced conformational changes¹⁶,

and multimerization-induced conformational changes¹⁷. Methods proposed to deconvolve sets of states when prior knowledge about one or more states is known include ablating residues corresponding to contacts of a known dominant state¹⁸ and supplementing the original multiple sequence alignment (MSA) with proteins known to occupy a rarer state¹⁹. However, there is a need for methods that deconvolve signal from multiple states if they are not already both present at the level of the entire MSA. For instance, simply subdividing a MSA and making predictions for portions of the MSA has also been used to detect variations in coevolutionary couplings within a protein family^{17,20}.

We hypothesized that metamorphic proteins, proteins that occupy more than one distinct secondary structure as part of their biological function²¹, would be a useful set of model proteins to develop methods for predicting conformational ensembles, as they undergo particularly striking conformational changes. For instance, although the metamorphic protein KaiB only contains 108 residues, it undergoes a conformational change that affects the secondary structure of roughly 40 residues in its C-terminal part, switching between a canonical thioredoxin-like structure and a unique alternate conformation (Figure 1a,b)⁵. Fewer than 10 metamorphic protein families have been thoroughly experimentally characterized²¹, which span a diverse range of functions. Fold-switching in proteins governs transcription regulation (RfaH in *E. coli*^{22,23}), circadian rhythms (KaiB in cyanobacteria⁵), enzymatic activity (the selegase metalloproteinase in *M. janaschii*²⁴), cell signaling (the chemokine lymphotactin in humans²⁵), and cell cycle checkpoints (Mad2 in humans²⁶⁻²⁸). A computational analysis of the PDB that identified changes in secondary structure between protein models sharing the same sequence suggested that between 0.5-4% of all proteins are fold-switching²⁹. The development of systematic methods to identify fold-switching proteins would aid in identifying new fold-switching proteins, highlight new structures and interactions to target for therapeutics²¹, as well as illuminate broader principles of protein structure, function, and evolutionary history that underlie known and as-of-yet undiscovered metamorphic proteins.

We hypothesized that if we could deconvolve sets of evolutionary couplings without adding prior knowledge and input these sets separately into AF2, AF2 might be able to predict multiple conformations with high structural accuracy. We demonstrate that a simple MSA subsampling method – clustering sequences by sequence similarity – allows AF2 to predict both states of the metamorphic proteins KaiB, RfaH, and Mad2. Importantly, we show that with our new method, “AF-Cluster”, both states are sampled and scored with high confidence by AF2’s learned predicted local Distance Difference test (plDDT) measure. We identified the reason for AF-Cluster’s prediction of multiple states: by making AF-Cluster predictions for KaiB variants from a curated phylogenetic tree, we found that KaiB variants predicted to fold to one or the other substate were distributed in clusters throughout the phylogenetic tree. We felt the strong need to experimentally test the AF-Cluster predictions on a new KaiB variant in *Thermosynechococcus elongatus vestitus* that was predicted to favor the fold-switched (FS) state. Using NMR, we could indeed verify our AF-Cluster prediction. To test the ability of our method to predict the effect of point mutations in switching a protein’s conformational equilibrium, we predicted and consequently validated a minimal set of point mutations that switch KaiB from *Rhodobacter sphaeroides* between the ground and FS state.

Having evaluated our AF-Cluster method on known metamorphic proteins, we next hypothesized that this approach might be able to detect alternate conformations in protein families for which no alternate structures are known. We applied our method to an existing database of MSAs associated with crystal structures³⁰. We describe here one candidate from our screen with a novel predicted alternate fold, the secreted oxidoreductase Mpt53 from *Mycobacterium tuberculosis*. Our results demonstrate that in the oncoming age of AF2-enabled structural biology, related sequences for any given protein target might harbor signal for more than one biologically-relevant structure, and that deep learning methods can be used to detect and analyze these multiple conformational states.

Results

AF-Cluster predicts both KaiB states.

We started our investigation with a contradiction posed by predicting the structure of the metamorphic protein KaiB using AF2. KaiB is a circadian rhythm protein found in cyanobacteria^{5,31} and proteobacteria³² that adopts two conformations with distinct secondary structures as part of its function: during the day, it primarily adopts the “ground state” conformation which has a secondary structure of $\beta\alpha\beta\beta\alpha\beta$ not found elsewhere in the PDB (**Figure 1a**, PDB: 2QKE). At night, it binds KaiC in a “fold-switch” (FS) conformation, which has a thioredoxin-like secondary structure ($\beta\alpha\beta\alpha\beta\alpha$) (**Figure 1b**, PDB: 5JYT). The thermodynamically favored state for KaiB from *Thermosynechococcus elongatus* (KaiB^{TE}) is the ground state; the FS structure was first solved in complex with KaiC³³, and could only be solved for the isolated KaiB by introducing stabilizing mutations to this variant³³. However, AF2 run using ColabFold³⁴ predicts the thermodynamically unfavored FS state for KaiB^{TE} (**Figure 1c, left**).

We hypothesized that evolutionary couplings present within the MSA may be biasing the prediction to the FS state. Interestingly, predicting the 3D structure of KaiB using just the 50 MSA sequences that are closest by number of mutations (henceforth referred to as edit distance) to KaiB^{TE} resulted in a prediction of the ground state (**Figure 1c, right**); however, predicting the 3D structure of KaiB^{TE} using the closest 100 sequences returned to predicting the FS state. Investigating this further revealed that the next 50 sequences themselves predicted the FS state in both AF2 and the unsupervised learning method MSA Transformer (**Extended Data Figure 1**). We thought that the MSA might contain subsets of sequences that yield AF2 predictions for either the ground or FS state, and that subsets that predicted the FS state would overpower subsets predicting the ground state. Therefore, we clustered the MSA by edit distance using DBSCAN³⁵, and ran AF2 predictions using these clusters as input (**Figure 1d**). We selected DBSCAN to perform clustering because we found it offered an automated route to optimizing clustering *a priori* (see Methods, **Extended Data Figure 2**). From here on we refer to

this entire pipeline as “AF-Cluster” – generating a MSA with ColabFold, clustering MSA sequences with DBSCAN, and running AF2 predictions for each cluster.

Strikingly, we found that the AF2 predictions from our MSA clusters comprised a distribution of structures, with the highest-scored regions of the distribution corresponding to the ground and FS state. **Figure 1e** depicts the top 5 models within 3 Å of crystal structures for each state, ranked by pLDDT. We compared this subsampling method to predictions from MSAs obtained by uniformly sampling over the MSA at various MSA sizes (**Figure 1f**), analogously to methods used elsewhere to predict multiple states of GPCRs¹⁴. We found that for uniformly subsampled MSAs of size 10, one of 500 samples was within 3 Å of the ground state, with lower confidence than the MSA cluster samples (**Extended Data Figure 2e**). Uniformly subsampled MSAs of size 100 did not sample the ground state at all.

We were curious if there were differing sets of contacts in our MSA clusters that other methods could also detect, and if this could help us understand how AF cluster detected two states. We used the same set of clusters to make predictions with the unsupervised deep learning model MSA Transformer³⁶, and found that these clusters contained evolutionary couplings for both states, and the score based on contact maps correlated with the RMSD in AF2 (see Methods, Extended Data Figure 3). No randomly-sampled MSAs were found to contain evolutionary couplings corresponding to the ground state.

Experimental test of KaiB predictions.

To better understand the origin of these two different sets of evolutionary couplings, we wanted to rule out the possibility that non-KaiB proteins with similar folds to the FS state were contributing to the prediction. We created a phylogenetic tree for KaiB that comprised 487 variants (see methods, **Extended Dataset 1**) and made structure predictions for all the variants. For each sequence, we used only the closest 10 sequences by evolutionary distance as an input MSA to best detect local differences in structure predictions. We found that regions of high pLDDT for both the ground and FS

state were interspersed across the tree (**Figure 2a**). We confirmed that for variants in the tree which had been experimentally characterized, the prediction from AF-Cluster corresponded with the structure expected to be thermodynamically favored (**Figure 2b**). For instance, variants from *Rhodobacter sphaeroides*³², *Thermosynechococcus elongatus*⁵, and *Synechococcus elongatus*³¹ all were predicted in the ground state, confirming their characterized circadian rhythm function. Conversely, a KaiB variant from *Legionella pneumophila* which has previously been crystallized in the FS state³⁷ was predicted with high confidence for the FS state.

KaiB variants in cyanobacteria have been characterized as belonging to three groups as well as a fourth variant, previously described as “elongated” KaiB due to a N-terminal domain of unknown homology and function³⁸. For clarity, we refer to the KaiB domain of this variant as KaiB-4. Intriguingly, we noticed that KaiB-4 variants were evolutionarily close to the more well-studied KaiB-1 variants involved in the circadian clock, yet the KaiB-4 variants were predicted by AF-Cluster to primarily occupy the FS state (**Figure 2c**). To experimentally test this prediction, we characterized one such variant with NMR spectroscopy, from *Thermosynechococcus elongatus vestitus* (henceforth referred to as KaiB^{TV}-4). KaiB^{TV}-4 was found to be stably folded at 35 °C and after backbone assignments we found peak duplication for many peaks corresponding to a major stable and minor unfolded state (**Extended Data Figure 4**). KaiB^{TV}-4 was confirmed to be monomeric at NMR concentration via size exclusion chromatography coupled to multi-angle light scattering (SEC-MALS) (**Extended Data Figure 4**). The secondary structure calculated from the major state chemical shifts indeed corresponded to the FS KaiB state (**Figure 2d**). CS-Rosetta³⁹ models calculated from the chemical shifts (**Figure 2e**) are within 1.8 ± 0.3 Å RMSD to the FS state predicted by AF-cluster. During revision we used ¹⁵N- and ¹³C-NOESY to determine the NMR structure as requested, and confirmed that the NMR structure (**Figure 2f**) also matches the AF-Cluster predicted model with 1.66 ± 0.3 Å RMSD, and an average pairwise RMSD of 0.6 Å over backbone atoms (**Extended Data Table 1**).

Mutations to flip KaiB's equilibrium.

Beyond predicting the predominant state of naturally-occurring proteins, we wished to test the ability of AF-Cluster in its ability to predict effects of point mutations, a task that AF2 in its default settings has not achieved⁴⁰. We hypothesized that by comparing clusters that predict different states, we could identify a minimal set of mutations that would switch AF2's prediction between states. We used KaiB from *Rhodobacter sphaeroides*³² (henceforth KaiB^{RS}) for this test, which we found by NMR switches between two monomeric states, to eliminate the complicating factor of mutations contributing to ground state tetramerization in the previously-studied KaiB^{TE}⁵. We observed that as for KaiB^{TE}, AF-Cluster predicts the ground and FS state for KaiB^{RS} with high confidence. We calculated the difference in enrichment between sequence clusters predicting the ground and FS state (**Figure 3a**), and noticed at several positions in the C-terminal part of the protein differentially enriched residues that differed substantially in their charge and hydrophobicity. For instance, clusters predicting the FS state were enriched for arginine at position 68, whereas clusters predicting the ground state at position 68 were enriched for leucine, a switch between a charged and a hydrophobic residue. We hypothesized that some subset of these mutations might be sufficient for determining whether AF2 predicts the ground or FS state. We folded all combination of the eight most-enriched residues in AF2 with no MSA to test if any combination caused a high-confidence fold-switch (see Methods, **Extended Data Figure 5**). Indeed, we found that three mutations: I68R, V83D, and N84R were sufficient to switch a prediction of KaiB^{RS} from the ground state to a prediction of the FS state (**Figure 3b**). We introduced these mutations into KaiB^{RS} and characterized this triple mutant (KaiB^{RS}-3m) with NMR (**Figure 3c**). It was again confirmed to be monomeric at NMR concentrations via SEC-MALS (**Extended Data Figure 3**). The ¹H ¹⁵N HSQC spectra in both WT and KaiB^{RS}-3m indicate the presence of major and minor state peaks, with the populations appearing to be flipped (**Figure 3d**). Strikingly, the secondary chemical shifts from backbone resonance assignment of the major peaks confirmed that the incorporation of these mutations indeed switch

KaiB^{RS} from the ground to the FS state (**Figure 3e**). Comparison of the average peak intensity ratios of the assignable minor (ground state) peaks to those of the major state (FS) peaks show that the mutant occupies the FS state with a population of 86% (versus 11% in WT), and ground state with a population of 14% (versus 89% in WT) (**Figure 3f**). Overall, NMR confirmed our prediction of a triple mutation to switch KaiB^{RS} to the FS state.

Testing AF-Cluster on other proteins.

We next tested AF-Cluster on five additional, experimentally verified fold-switching proteins: the *E. coli* transcription and translation factor RfaH, the human cell cycle checkpoint Mad2, the selease metalloproteinase enzyme from *M. janaschii*, the human cytokine lymphotactin, and the human chloride channel CLIC1. In RfaH, the C-terminal domain (CTD) interconverts between an α -helix bundle and a β -barrel through binding to functional partners²³. In the autoinhibited state, the α -helix bundle of the CTD interacts with the NTD. In the active state, the CTD unbinds and forms a β -barrel (**Figure 4a**)^{22,23}. Predicting the structure of RfaH with the complete MSA from ColabFold returned a structure that largely matched the autoinhibited state (**Extended Data Figure 6a**) apart from the first helical turn in the CTD being predicted as disordered. We note that the B-factors in the crystal structure for this region are the highest (**Extended Data Figure 6b**). The active state was not predicted. In contrast, AF-Cluster predicted both the autoinhibited and the active state (**Figure 4b**). Notably, the average pLDDT for the top 5 models for each state (84.2 for the active state, 73.9 for the autoinhibited) was higher than the pLDDT of the autoinhibited state by the complete MSA (pLDDT of 68.6), suggesting that clustering resulted in deconvolving conflicting sets of couplings.

Mad2 has two topologically distinct monomeric structures that are in equilibrium under physiological conditions²⁷. These are termed the open and closed states (often referred to as O-Mad2 and C-Mad2). The closed state binds Cdc20 as part of Mad2's function as a cell cycle checkpoint²⁶. In the closed state, the C-terminal beta-hairpin rearrange into a new β -hairpin that binds to a completely

different site, displacing the original N-terminal β -strand²⁸ (**Figure 4c**). We found AF-Cluster was again capable of predicting models for both of Mad2's conformational states (**Figure 4d**).

RfaH and Mad2 both interconvert between two distinct monomeric forms. However, selease, lymphotactin, and CLIC1 both interconvert between a monomeric and an oligomeric state (**Extended Data Figure 6c**). AF-Cluster was unable to predict the oligomeric state for selease, lymphotactin, and CLIC1. The selease protein is a metallopeptidase from *M. janaschii* first reported by Lopez-Pelegrin et al²⁴. It reversibly interconverts between an active monomeric form and inactive dimers and tetramers. Lymphotactin is a human cytokine that adopts a cytokine-like fold but was found to adopt an all- β -sheet dimer via NMR at higher temperature and in the absence of salt²⁵. CLIC1 is an ion channel with a redox-enabled conformational switch. In the reduced state, it adopts a monomeric state with a N-terminal $\beta\alpha\beta\alpha\beta$ fold. Upon being oxidized, it forms a dimer, and its N-terminus adopts a $\alpha\alpha\alpha$ fold. This fold is stabilized by a disulfide bond between two of the α -helices within the monomer that forms upon oxidation⁴¹. All these proteins pose starting points for future improvements to AF-Cluster.

AF-Cluster predicts novel states.

We next asked if AF-Cluster could detect novel putative alternate states in protein families without known fold-switching (**Figure 5a**). As a starting point, we selected 628 proteins with length 48-150 amino acids from a database of MSAs associated with crystal structures³⁰ (see Methods). After clustering the MSAs using DBSCAN³⁵, we generated AF2 predictions for 10 randomly-chosen clusters from each family and compared the pLDDT to the RMSD from the reference structure. For most of the protein families screened, an increase in RMSD corresponded to a decrease in pLDDT (**Figure 5b**). As a control, AF-Cluster models of Ubiquitin, a protein well-characterized to have no alternate states, only returns models with high confidence and low RMSD to the crystal structure 1UBQ. However, a handful of proteins in this preliminary screen returned models with high RMSD and high pLDDT, hinting to a

predicted structure with high dissimilarity to the original structure as well as high confidence from AF2.

For these proteins, we generated AF2 predictions for all generated clusters from the MSA.

Results for one of these fold-switching candidates, the oxidoreductase Mpt53 from *M. tuberculosis*, is described here. Mpt53 is an extracellular single domain enzyme that is suggested to ensure correct folding of several cell-wall and extracellular protein substrates in *M. tuberculosis* by catalyzing disulfide oxidation⁴². **Figure 5c** depicts all the AF-Cluster models for Mpt53, visualized by principal component analysis (PCA) on the set of closest heavy-atom contact distances. Two prominent states are observed that correspond to the largest-sized MSA clusters (**Extended Data Figure 7a**), and both of which have pLDDT values statistically significantly higher than the rest of the set (**Extended Data Figure 7b**). One state corresponds to the known thioredoxin-like conformation of Mpt53⁴², whereas the other state corresponds to a novel conformation with a different secondary structure layout (**Figure 5d,e**). In the second state, strand $\beta 1$ replaces $\beta 5$ within the β -sheet. The α -helix $\alpha 4$ is displaced to the opposite side of the β -sheet, and $\alpha 5$ is rotated. Mpt53 is a member of a superfamily of enzymes with diverse functions that all share the same thioredoxin fold with a conserved CxxC active site that can form a disulfide bond. Models for the alternate state demonstrate a very similar active site orientation at residues C36-C39 (**Extended Data Figure 7c**). We were curious if we could find structures in the PDB that matched this alternate state. We screened for homologous 3D structures for both 1LU4 and the alternate state in the PDB using DALI⁴³ (see Methods, **Extended Data Figure 7d-f**). The closest structure we found (3EMX) adopted a similar secondary structure to the Mpt53 alternate structure. This structure is of an unspecified thioredoxin from the archaea *Aeropyrum pernix* with no associated publication.

We were curious if any structure homologues to the known Mpt53 state also predicted alternate conformations. We used AF-Cluster to test 10 proteins with the lowest alignment-weighted RMSD from DALI to the original state (see Methods). Notably, 6 of the 10 sampled an analogous alternate fold with varying amounts of sampling (**Extended Data Figure 8**). The closest-ranked

homologues for both the known and alternate state are dispersed across a calculated phylogenetic tree of all the DALI hits (**Extended Data Figure 9**).

Discussion

AlphaFold2 (AF2) has revolutionized prediction of single structures⁴⁴, but devising methods to predict structures of multiple conformational states would significantly advance our understanding of protein function at atomic resolution. We demonstrate that simply clustering input sequences from MSAs of metamorphic proteins enables AF2 to sample multiple biologically relevant conformations with high confidence.

Using the metamorphic protein KaiB as a model system, we sought to understand why clustering resulted in multiple states predicted. We found that pockets of KaiB variants in a phylogenetic tree were predicted to be stabilized for one or the other state. This is consistent with findings for the fold-switchers RfaH⁴⁵ and lympotactin⁴⁶, as well as non-fold-switching proteins like the Cro repressor family⁴⁷. However, the myriad roles of KaiB in bacteria have yet to be fully understood: some bacteria contain up to 4 copies of KaiB, only one of which has been extensively studied³⁸. One KaiB variant in *Legionella pneumophila*, which was crystallized in the FS state, was found to not be involved in circadian rhythms, but instead implicated in stress responses³⁷. We identified a KaiB variant in *T. elongatus vestitus* that is phylogenetically close to the known fold-switching KaiB for which the ground state is thermodynamically favored, yet was predicted and experimentally corroborated to be stabilized for the FS state. Intriguingly, predicting this variant in single-sequence mode in AF2 incorrectly predicts the ground state (see Supplemental Discussion), further underscoring the utility of isolating local evolutionary couplings by clustering sequences. Our findings open new biological questions to identify the biological role of this KaiB copy in cyanobacteria in the future.

However, considering that an ideal sampler would sample and score models in accordance with an underlying Boltzmann distribution, this method suffers from several limitations. Firstly, the pLDDT

metric itself cannot be used as a measure of free energy. This was immediately evident in our investigation of KaiB, where in our models generated with AF-Cluster, the thermodynamically-disfavored FS state still had higher pLDDT than the ground state (**Extended Data Fig. 2e**). Furthermore, increasing evidence indicates that low pLDDT correlates with regions with high local disorder as measured by backbone order parameters⁴⁸. Secondly, the number of models returned for each state from AF-Cluster will reflect the abundance of constructs reflecting different states across the protein family, which cannot be interpreted as that state's Boltzmann weight. We tested other methods for introducing noise in AF2 using KaiB^{RS} with no MSA as a test: sampling across the 5 models, incorporating dropout, and using random seeds, and found none of these to cause AF2 to predict any models of the FS state (see Supplemental Discussion).

Disease-causing point mutations are often due to population changes of protein substates^{3,4}, and thus there is great interest for methods to predict the effects of point mutations on structural ensembles and free energy. We found that the information provided by our new AF-Cluster method was sufficiently predictive to inform the design of 3 point mutations that could switch the equilibrium of KaiB^{RS} from the ground to FS state. This work also establishes the KaiB^{RS} variant as a facile system for testing multistate design and thermodynamic prediction methods.

Though our design of KaiB was performed using AF-Cluster with no MSA, we were curious if AF-Cluster's sensitivity to the effects of point mutations could be generalized to other systems where single point mutations have been demonstrated to completely switch folds. We tested 12 sets of point mutations in the G_A/G_B protein system. Starting from two naturally occurring 56 amino acid domains from the multidomain protein G, in which G_A adopts a 3-alpha-helix and G_B a 4b+a fold, variants had been engineered to switch between both folds⁴⁹⁻⁵¹ (**Extended Data Figure 10**). Unlike the point mutations in KaiB, which were selected from evolutionary sequence abundances, these were

engineered via selection of extensive variants. We found that the highest-pLDDT model from AF-Cluster correctly predicted the most stable folds for 10/12, whereas default AF2 correctly predicted 8/12.

By using AF-Cluster to screen protein families not known to fold-switch into alternate states, we identified a putative alternate state for the oxidoreductase Mpt53 in *Mycobacterium tuberculosis*. Mpt53 oxidizes the human kinase TAK1, which was shown to trigger immune response⁵². The thioredoxin superfamily containing Mpt53 is a ubiquitous set of enzymes known for their promiscuous catalytic activity, being able to reduce, oxidize, and isomerize disulfide bonds⁵³. Theoretical work suggests that conformational change is the most parsimonious explanation of the evolution of promiscuous activity in the thioredoxin family⁵⁴. Given that known metamorphic proteins often switch folds through cellular stimuli, it may in general be difficult to experimentally validate novel folds identified through computational methods if the stimulus – whether pH, redox reaction, a binding partner – is unknown.

We speculate that there may be many more uncharacterized functional states of proteins present that this method could uncover. The AlphaFold protein structure prediction database⁵⁵ contained 214 million predictions of single structures as of June 2023. If Porter and Looger's estimate²⁹ that 0.5-4% of all proteins contain fold-switching domains is accurate, this would correspond to roughly 1-8 million fold-switching proteins with possible alternate states that would not be predicted by the default AF2 method.

Further study is ongoing in what types of conformational changes AF-Cluster and other methods based on altering input MSAs can predict. Because previous works have identified coevolutionary couplings corresponding to multiple states of domain-based conformational changes^{15,16,20}, we speculate that clustering-based MSA preprocessing methods will offer improvements over existing methods¹⁴, and importantly, insights in the evolution of multiple conformational states. However, conformational substates not present in coevolutionary signal may require alternate methods. All methods also need to

be evaluated and improved in their ability to sample and score in accordance with the system's underlying Boltzmann distribution. As protein sequencing data continues to increase, computational methods for characterizing and identifying conformational substates will likely provide increasing insight into protein folding, allostery, and function.

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the paper. H.K.W.-S., A.O., J.A., R.O., S.O., L.C., and D.K. commented on the manuscript and contributed to data interpretation.

Figure legends

Figure 1: AF2 predictions from MSA clusters for the fold-switching protein KaiB return both known structures. Crystal structures of KaiB of (a) the “ground” state from *T. elongatus* (KaiB^{TE}) (PDB: 2QKE) and (b) the “fold-switch” (FS) state (PDB: 5JYT). c) The default ColabFold prediction of KaiB^{TE} returns the FS state. Using only the closest 50 sequences by sequence distance returned from the MSA returns the ground state. (a-c) The first 50 residues that are identical in both states are colored grey and the fold-switching elements colored by the same color in both states. d) Overview of “AF-Cluster” method. Left: MSA is clustered by sequence similarity. Sequence space is depicted using a tSNE embedding⁵⁸ of the one-hot sequence encoding. Right: clusters are used as input to AlphaFold2, resulting in a distribution of predicted structures, colored by pLDDT. e) Top 5 models for the ground and FS state, ranked by pLDDT. f) RMSD of AF2 structure predictions for all clusters relative to the ground and FS state. The highest-confidence regions of the AF-Cluster distribution for KaiB^{TE} are within 3 Å RMSD of crystal structures of both the ground and FS state. In contrast, sampling the MSA uniformly returns only the FS state with high confidence.

Figure 2: The KaiB family contains pockets of sequences predicted to be stabilized for both states. a) AF2 predictions for each variant in a phylogenetic tree⁸⁹ using the 10 closest sequences as input MSA. Left: Each node is colored by predicted state (blue: ground state, red: FS state). Right: Same tree, colored by pLDDT. b) Three known fold-switching KaiB variants from i) *Rhodobacter sphaeroides*³², ii) *Themosynechococcus elongatus*⁵, and iii) *Synechococcus elongatus*³¹, are predicted in the ground state, and a variant from iv) *Legionella pneumonia*³⁷, crystallized in the FS state, is predicted in the FS state with high pLDDT. c) A KaiB copy present in *T. elongatus*, KaiB^{TV}-4, is predicted to favor the FS state. d-f) Experimental testing of KaiB^{TV}-4: d) Secondary structure propensity determined by NMR backbone chemical shifts, calculated with TALOS-N⁷¹ for KaiB^{TV}-4, fully agrees with the FS state predicted by AF-Cluster. Unassigned amino acid residues are indicated by an asterisk. e) Structure models calculated with CS-Rosetta³⁹, shown in grey, have 1.8 ± 0.3 Å RMSD to the AF-Cluster model (in magenta). f) NMR structural models calculated from ¹⁵N and ¹³C edited NOESY spectra have an average pairwise RMSD of 0.6 Å, and 1.66 ± 0.3 Å RMSD to the AF-Cluster model. RMSD values in (e) and (f) are calculated over backbone atoms in secondary structure regions.

Figure 3: A designed minimal set of mutations switches the predominant fold of KaiB from *Rhodobacter sphaeroides* (KaiB^{RS}) from the ground to the FS switched state. a) Sequence features enriched in clusters that predict the FS and ground state. b) Three mutations are sufficient to switch the structure prediction for KaiB^{RS} in AF2 from the ground state to the FS state. Top: AF-Cluster models for KaiB^{RS} and KaiB^{RS}-3m, colored by pLDDT. Bottom: 3 mutation sites are highlighted. c) Overlaid ¹H¹⁵N HSQC spectra of KaiB^{RS} (blue) and KaiB^{RS}-3m (red). d) Examples of residues from well-resolved regions in the ¹H¹⁵N HSQC assigned in both states are shown for wild-type KaiB^{RS} and KaiB^{RS}-3m to illustrate the flip in populations via the 3 mutations. e) Chemical shift-based secondary structure calculated with TALOS-N⁷¹ of the ground and FS states of KaiB^{RS} and the major state of KaiB^{RS}-3mut. Unassigned amino acid residues are indicated by an asterisk. Rectangular box in green indicate fold-switching region. f) Average of the NMR peak intensity ratio of ground versus FS state for select residues that could be assigned in both states for both variants in well-resolved regions. Error bars represent standard error (n=5 residues).

Figure 4: AF-Cluster predicts fold-switching for the proteins RfaH and Mad2. a) Fold-switching in the RfaH transcription factor in *E. coli*. In RfaH's autoinhibited state, the C-terminal domain (CTD, red) forms an alpha-helix bundle (PDB: 5OND)⁹⁰. In the active state, the CTD unbinds and forms a β -sheet that is homologous to the transcription factor NusG (CTD PDB: 2LCL)²². b) AF-Cluster returns structure models that include both the autoinhibited and the active state with high confidence. Note that the C-terminal domain orientation is of course not defined due to the flexible linker between the two domains. c) The closed state (PDB: 1S2H)²⁷ and the open state (PDB: 1DUJ)⁹¹ of the Mad-2 spindle checkpoint in humans with the fold-switching portions colored. d) Both Mad-2 states are predicted by AF-Cluster with high confidence.

Figure 5: Screening for fold-switching in many protein families predicts a putative alternate fold for *M. tuberculosis* secreted protein Mpt53. a) Strategy for detecting novel predicted alternate folds. Screening of 628 families with more than 1000 sequences in their MSA and residue length 48-150 from ref.³⁰. After clustering, we ran AF2 predictions from 10 randomly-selected clusters from each. b) Candidates for further sampling were selected by looking for outlier predictions with high RMSD to the reference structure and high pLDDT. c) Sampled models for candidate Mpt53, visualized with a principal component analysis on closest heavy-atom contacts. Two states with higher pLDDT than background are observed. d) Top 5 models by pLDDT for the known state (top) and the putative alternate state (bottom), colored by pLDDT per residue. e) Top: Crystal structure of reduced state of Mtb Mpt53 (PDB: 1LU4), which corresponds to state 1 in the sampled landscape. In the putative alternate state 2, strand β 1 replaces β 5 in the 5-strand β -sheet. Helix α 4 shifts to the other side of the β -sheet and helix α 5 is displaced.

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Methods

MSA generation. Multiple sequence alignments (MSAs) were generated using the MMseqs2⁵⁶-based routine implemented in ColabFold³⁴. In brief, the ColabFold MSA generation routine searches the query sequence in three iterations against consensus sequences from the UniRef30 database⁵⁷. Hits are accepted with an E-value lower than 0.1. For each hit, its respective UniRef100 cluster member is realigned to the profile generated in the last iterative search, filtered such that no cluster has higher max sequence identity than 95%, and added to the MSA. Additionally, in the last round of MSA construction, sequences are filtered to keep the 3000 most diverse sequences in the sequence identity buckets [0.0–0.2], (0.2–0.4], (0.4–0.6], (0.6–0.8] and (0.8–1.0].³⁴ Prior to clustering, we removed sequences from the MSA containing more than 25% gaps.

Clustering. We found that our method for parameter selection in DBSCAN³⁵ empirically optimized predicting KaiB's two states from its MSA with no prior information about the KaiB landscape in the following way. An optimal clustering to identify sets of contacts corresponding to multiple states needs to balance two size effects: if clusters are too small, they may contain insufficient signal to capture any state. However, if clusters are too large, they may dilute signal from some states, an extreme case of this exemplified in how KaiB predicted with its entire MSA resulted in only the FS state. In brief, DBSCAN³⁵ clusters datapoints by identifying "core" density regions where at least k points fall within distance ϵ from one another. Points farther than ϵ from points in core density regions are excluded as noise. Clustering the KaiB MSA with varying ϵ values resulted in a peak in the number of clusters returned (**Extended Data Figure 2a**). We termed the ϵ corresponding to this peak $\epsilon^{\max}$. For $\epsilon < \epsilon^{\max}$, the number of clusters is lower because more sequences are left unclustered as outliers (**Extended Data Figure 2b**). For $\epsilon > \epsilon^{\max}$, more sequences are clustered, so the number of clusters is decreasing because clusters are merged.

We investigated the effect of varying epsilon on resulting AF2 predictions for the protein KaiB. **Extended Data Figure 2c** depicts clusters in sequence space (represented with tSNE⁵⁸ on sequence one-hot encoding), and **Extended Data Figure 2d** depicts the structure landscape of these clusters. Epsilon was varied between 3 and 20 with step size 0.5. For the preliminary scan of 628 protein families, this sweep on epsilon was performed on a randomly-selected 25% of the MSA to accelerate computation.

Investigating evolutionary couplings from clustering with MSA Transformer. We wished to probe the degree and nature of evolutionary couplings in clusterings from the AF-Cluster method and compare them to clusterings from random sampling. To do this, we made predictions for DBSCAN-generated KaiB clusters in the model MSA Transformer³⁶ using its default settings. MSA transformer is an unsupervised learning method, which signifies that its contact predictions purely reflect evolutionary couplings learned in sequences, rather than being supervised on structure as is the case AF2. For clusters with more than 128 sequences, the default “greedy subsampling” routine was used to select sequences.

We compared clusters sampled with both AF-Cluster (329 samples) and randomly sampled with size 10 and 100 (500 samples each). We scored predicted contact maps to the KaiB ground and FS state using a standard Area Under the Curve (AUC) metric assessing the accuracy of a fraction of top k predicted contacts that are correct for $k=1$ up to L , where L is the length of the protein³⁶. Every cluster was thus assigned a corresponding “Ground state AUC” and “FS state AUC” reflecting its similarity to both states. Contact maps for both states used in this scoring are depicted in **Extended Data Figure 3a**.

We found that clusters from AF-Cluster scored higher to the Ground state (**Extended Data Figure 3b**), and that the highest-scoring randomly-sampled cluster did not contain the secondary structure feature most emblematic of the ground state: the C-terminal beta-strand (boxed in **Extended Data Figure 3ci**, absent from 2cii.) For both states, we found that the AUC scores correlated with the RMSD to the FS state from AF2 (Ground state: Spearman $R = -0.32$, $p=2e-09$, FS state: Spearman $R = -$

0.34, $p=4e-10$), suggesting that the evolutionary couplings that MSA Transformer detected in each cluster also affected predictions in AF2.

Phylogenetic tree construction. A candidate set of sequences was identified using BLASTP v 2.6.0⁵⁹ using the protein sequence for KaiB from *Synechococcus elongatus* (WP_011242647.1) as a query. The query was run against the NCBI non-redundant protein database with the exclusion of models or uncultured/environmental sample sequences. The selected 1270 sequences were aligned using MAFFT⁶⁰. The alignment was used to generate an untrimmed phylogenetic tree in RAxML (version 8.2.9)⁶¹. Next, the alignment was trimmed down to only include sequences with sequence homology of 90% or less using CD-HIT⁶². In addition, sequences that showed excessive length compared to the search input were removed or, if possible, trimmed to only reflect the KaiB domain. We selected sequences to ensure coverage for the different clades based on the original, large RAxML tree. Lastly, this was cross-checked with a full KaiC tree published in ref. ³² to ensure coverage across all phyla expected to contain KaiB-type proteins. For the calculation of the final phylogenetic tree, the curated set of 487 sequences was aligned with MAFFT⁶⁰ using the E-INS-I algorithm (see **Extended Dataset 1**). This alignment was then used as input for PAML (version 3.3.20170116)⁶³ to create a KaiB phylogenetic tree. The “LG” model was applied with twelve substitution rate categories^{64,65} and the tree topology, branch lengths, and the substitution model parameters were optimized. This resulted in the final tree used in this manuscript (see **Extended Dataset 1**).

Protein expression and purification. The KaiB domain of KaiB^{TV}-4 (WP_011056401.1) and wild-type KaiB^{RS} (WP_002725098.1) constructs were ordered from GenScript (**Supplementary Table 1**). The plasmid was subcloned into the Nco1 and Kpn1 sites of the pETM-41 vector. The triple mutant (I68R/V83D/N84A) of KaiB^{RS} used in this study was generated by following the Q5 Site Directed Mutagenesis protocol using wild-type KaiB^{RS} as template. All primers were ordered from GeneWiz (New

England Biolabs) (**Supplementary Table 1**). The triple mutation was confirmed by DNA sequencing using GeneWiz primers.

The pETM-41 plasmids encoding for wildtype KaiB^{RS}, triple mutant KaiB^{RS}, and KaiB^{TV}-4 were transformed into *Escherichia coli* BL21(DE3) cells (New England Biolabs). To prepare ¹³C-¹⁵N isotopically-labeled samples for NMR studies, three colonies selected from a freshly transformed plate containing 50 µg/ml Kanamycin (Kan) were used to inoculate 10 ml each of LB^{Kan} cultures. The LB starter cultures were grown for 6 hours at 37°C with shaking at 220 rpm. The LB starter cultures were combined and used to inoculate an overnight minimal (M9) starter culture with a starting OD₆₀₀ of 0.002. 1 Liter M9 media supplemented with 1 g/l of ¹⁵NH₄Cl and 2 g/l of ¹³C₆ glucose was inoculated using 25 ml of overnight M9 culture, then grown to OD₆₀₀ of 0.7 at 37°C before inducing with 0.5 mM Isopropyl β-d-1-thiogalactopyranoside (IPTG) at 21°C. This culture was grown overnight with shaking at 220 rpm.

KaiB^{RS} and KaiB^{TV}-4 were purified using similar method as previously described for KaiB^{RS} ³². Briefly, cell pellets were resuspended in lysis buffer containing 50 mM Tris pH 7.5, 250 mM NaCl, 2 mM TCEP, 10% glycerol, 10 mM Imidazole, 1X EDTA-free protease inhibitor cocktail (Thermo Fisher Scientific), DNase I (Sigma Aldrich), and lysozyme (Sigma Aldrich). Lysate was sonicated on ice for 15 min (20 s on, 30 s off, output power of 40W), followed by centrifugation at 18500 rpm for 45 min at 4°C. The supernatant was filtered before loading on HisPur nickel metal-chelated agarose beads (Thermo Fisher Scientific) pre-equilibrated with Buffer A (50 mM Tris pH 7.5, 250 mM NaCl, 2 mM TCEP, 10% glycerol, 10 mM Imidazole). The resin was washed with Buffer A, followed by further removal of impurities using 5-15% Buffer B (50 mM Tris pH 7.5, 250 mM NaCl, 2 mM TCEP, 10% glycerol, 500 mM Imidazole) in a stepwise manner. The proteins eluted at 50 % Buffer B. The eluted proteins were cleaved with TEV protease to remove His₆-MBP tag from KaiB^{RS} and KaiB^{TV}-4 during overnight dialysis in 50 mM Tris pH 7.5, 250 mM NaCl, 2 mM TCEP, 10% glycerol. Cleaved samples were reloaded on HisPur nickel metal-chelated agarose beads to collect cleaved KaiB^{RS} and KaiB^{TV}-4. Cleaved samples were further purified on

S75 size-exclusion chromatography column in 100 mM MOPS, pH 6.5, 50 mM NaCl, 2 mM TCEP for NMR studies. All samples were purified to homogeneity with a single band at ~10 kDa on a Bis-Tris 4–12% gradient SDS–PAGE gel (GenScript). The protein concentration was determined using microplate BCA protein assay kit (Thermo Fisher Scientific). The yield for KaiB^{RS} triple mutant was ~22 mg per 1L cell culture and ~6 g per 1L cell culture for KaiB^{TV}-4. ¹³C-¹⁵N KaiB^{RS}-3m and KaiB^{TV}-4 NMR samples used for data collection were 1.8 mM (~300 µl) and 1.1 mM (~200 µl) respectively in 100 mM MOPS, pH 6.5, 50 mM NaCl, 2 mM TCEP, 10% D₂O. Samples used for NMR data collection were enclosed 5-mm susceptibility-matched Shigemi NMR tube for ¹⁵N KaiB^{RS}-3m and wildtype or a 3-mm NMR tube for KaiB^{TV}-4.

NMR data collection and processing. NMR data were collected at 293°K and 308°K for KaiB^{RS}, and at 308°K KaiB^{TV}-4 on Varian VNMRS DD 800-MHz or Bruker Avance III HD 750-MHz with a triple-resonance TXI Cryoprobe or Avance NEO 800 spectrometer equipped with a triple-resonance TCI Cryoprobe or Varian VNMRS DD 600-MHz equipped with a triple resonance cold probe. All experiments were run by using the Varian VnmrJ software library (VnmrJ version 4.2 software from Varian, Inc.). All 3D spectra for KaiB^{RS}-3m and KaiB^{TV}-4 were recorded using non-uniform sampling (NUS) with a sampling rate of ~30% and standard sampling for KaiB^{RS} wildtype. Backbone ¹³C, ¹⁵N, H^N resonance assignments were performed using standard double and triple resonance experiments (¹H, ¹⁵N-HSQC, HNCACB, CBCA(CO)NH, HNCOCa, and HNCA). All NMR data were processed using NmrPipe⁶⁶, and the NUS data were reconstructed and processed using the SMILE⁶⁷ package, included with NMRPipe⁶⁶.

NMR Data Analysis and structure calculation. Backbone resonances were assigned in the POKY⁶⁸ software package using 2D ¹H, ¹⁵N HSQC, 3D HNCACB, CBCA(CO)NH, HNCOCa, and HNCA spectra. The peaks were initially picked using the APES tool in POKY⁶⁸ and verified manually, followed by peak lists submission to I-PINE⁶⁹ web server via the PINE-PARKY.2⁷⁰ plugin in POKY for automated assignments of the backbone resonances. The assignments from I-PINE were verified and some adjusted manually in

POKY. The side-chain atoms of KaiB^{TV}-4 were manually assigned using 2D ¹H-¹³C HSQC (aliphatic) and ¹H-¹³C HSQC (aromatic), 3D HBHA(CO)NH, HCCH-TOCSY (aliphatic), HCCH-TOCSY (aromatic), C(CO)NH, H(CCO)NH, 2D (HB)CB(CGCD)HD (aromatic), and 2D (HB)CB(CGCDCE)HDHE (aromatic) spectra. Secondary structure propensities were calculated using TALOS-N⁷¹. CS-Rosetta³⁹ structure models were calculated within the I-PINE webserver by submitting a manually curated peak list corresponding to the major folded state. Average peak intensity ratios were determined by selecting 5 amino acid residues that had both ground state and FS state peaks assigned in WT KaiB^{RS} and KaiB^{RS}-3m from well resolved regions in the ¹⁵N-HSQC spectra.

The solution NMR structure of ¹³C, ¹⁵N labeled KaiB^{TV}-4 was solved using the Integrative NMR⁷² package via POKY. 3D ¹H, ¹⁵N HSQC NOESY, ¹H, ¹³C HSQC NOESY (aliphatic), and ¹H, ¹³C HSQC NOESY (aromatic) were used in addition to backbone and sidechain resonance assignments for structure calculation. Peak lists were generated using either the APES tool or iPick (integrated UCSF peak picker) in POKY, followed by manual inspection of peaks. X-PLOR-NIH⁷³-based calculations were used for all steps of structure calculations and refinement in the PONDEROSA C/S package⁷⁴. Firstly, several unambiguous NOEs were assigned manually including those that already defined the beta-strand topology unique to the FS state (strip plot and diagram in **Extended Data Figure 4b,c**). We followed this with automated NOE assignments by AUDANA⁷⁵ (which employs X-PLOR-NIH for simulated annealing and TALOS-N for calculation torsion angle constraints). For the AUDANA automation steps, our predicted model of KaiB^{TV}-4 was used as a structural starting point (Fig. 2d). Generated distance constraints from AUDANA were carefully validated using the PONDEROSA Analyzer interfaced with the PONDEROSA Connector tool in the POKY and PyMOL⁷⁶ software. A white-list/black-list was also generated in the PONDEROSA analyzer and used as restraints to aid efficient NOE assignment in the subsequent round of AUDANA run. Using the NOE distance constraint files generated from AUDANA, Constraints-Only X-plor NIH calculations were performed in iterative cycles to refine the NOE distances. In this step, 40 structures are calculated,

and of these 20 lowest energy structures were used in the Final Step of refinement. We finalized the constraint refinement by running a final step with explicit water refinement. This step provided 20 out of 200 lowest energy structures and performed energy minimization in a water box. The final structures were validated using the wwPDB validation tool^{77,78} (<https://validate-rcsb-east.wwpdb.org/validservice/>) and the Protein Structure Validation Suite (PSVS)⁷⁹. Based on Procheck⁸⁰ analysis on secondary structure elements, the Ramachandran statistics among the top 20 lowest energy structures are 98% for most favored regions, 2% for additional allowed regions, and 0% for disallowed regions. The structure calculation statistics for the 20 lowest energy structures are in **Extended Data Table 1**. All NMR-related software for assignments and structure calculations was accessed in NMRbox⁸¹.

Size Exclusion Chromatography coupled to Multi Angle Light Scattering (SEC-MALS). To determine the oligomeric state of KaiB^{RS}-3m and KaiB^{TV}-4, 100 μ l of 500 μ M purified protein was loaded onto a Superdex 75 increase 10/300 GL column (Cytiva) equilibrated at 0.25 ml/min flow rate (AKTA HPLC system) (Extended Data Figure 4) in 100 mM MOPS, pH 6.5, 50 mM NaCl, 2 mM TCEP. Detection was performed using a MiniDAWN multi angle light scattering detector and an Optilab differential refractometer (Wyatt Technology). Molecular weights were calculated with the Astra 8.1.2.1 software using a differential index of refraction (dn/dc) value of 0.185 ml/g.

Data selection for fold-switching screening. Protein families were selected from a database previously developed to query the origins of spatially distant coevolutionary contacts³⁰. The database consisted of nonredundant proteins with associated X-ray structures with resolution < 2 Å. The MSAs were originally constructed using HHblits⁸² run against the UniProt database and filtered to exclude sequences with high similarity³⁰. Though the database originally contained 9,846 proteins, for this preliminary work we only selected proteins with sequence length between 52 and 150 residues and with more than 1000 sequences in the alignment, which totaled 628 proteins.

Screening for Mpt53 structure homologues. We used DALI⁴³ to screen for structure homologues to both the known and putative alternate Mpt53 structure. We used the DALI webserver to search the PDB (<http://ekhidna2.biocenter.helsinki.fi/dali/>) and downloaded all PDB hits. We filtered both sets of hits for unique sequences as well as unique models, i.e., to retain just one chain per model if multiple chains were returned. This resulted in 1822 matches for the Mpt53 known state and 1245 matches for the Mpt53 alternate state (**Extended Data Figure 7d**). We took the union of these two sets and applied CD-HIT⁶² with default parameters to filter for highly similar sequences. This resulted in 1055 sequences remaining. 479 of these were hits for both the known and alternate state, with 368 exclusively for the known and 208 exclusively for the alternate state.

To identify matches with the best RMSD considering the length of the alignment, we calculated the “weighted RMSD” as

$$\text{weighted RMSD} = \frac{\text{RMSD}}{\text{fraction aligned'}}$$

where the fraction aligned is the alignment length returned by DALI divided by the total length of the sequence in the matching structure. We observed that the matches exclusively for one or the other state had worse weighted RMSD for their structure compared to matches that matched both structures (**Extended Data Figure 7d**), and therefore focused our analysis on the 479 structures that matched both states. The weighted RMSD for both states for these are plotted in **Extended Data Figure 7e**.

A few structures had higher weighted RMSD for the alternate Mpt53 state than for the known Mpt53 state (orange points in **Extended Data Figure 7e**, structures in **Extended Data Figure 7f**). 7 of the depicted 9 proteins had a helix in an analogous spot to the alpha-4 helix location in the Mpt53 alternate structure. One structure, 3EMX, had a N-terminal beta-strand arranged in the same conformation as the Mpt53 alternate state. Deposition data for these structures are given in **Supplementary Table 2**.

To test if these sequences had any phylogenetic similarity, we took the 1055 sequences representing the union of both sets of matches, filtered for sequence length less than 500, and aligned using the MAFFT⁸³ webserver with default parameters. We calculated a phylogenetic tree using IQ-TREE⁸⁴ with the LG+I+G substitution model. The resulting tree is depicted in **Extended Data Figure 9**, and demonstrates that while the closest structure homologues to the known state are clustered, the closest homologues to the alternate state are dispersed across the tree.

Testing the sensitivity of AF2 and AF-Cluster to point mutations in the G_A/G_B ⁸⁵ system. MSAs were generated using the default MSA generation routine from ColabFold, using MMseqs2. For AF-Cluster, MSAs were then clustered using the DBSCAN procedure as described earlier. MSAs were used as input to AF2 run in all 5 models with 0 recycles and 8 random seeds. Sequences of the 12 point mutation sets are depicted in **Extended Data Figure 10a**. A representative clustering for variant GA98 is depicted in **Extended Data Figure 10b**. Investigating a few sequences from each cluster revealed that sequences of different lengths corresponded to G_B -like and G_A -like proteins.

For each point mutant, we compared models generated with the default MSA, AF-Cluster MSAs, and an MSA from both the wild-type G_A and G_B variant reported in ref.⁵⁰. The TMscores of resulting models and their pLDDTs are plotted in Extended Data Figure 10c. For 4/12 point mutants, the default ColabFold MSA failed to return any models corresponding to the correct structure. AF-Cluster corrected two of these – G_B 95 and G_B 88. For the remaining 2 that AF-Cluster failed to predict, using the wildtype G_B MSA returns a higher-scoring model than the wildtype G_A MSA, suggesting the limitation is in either the sequence retrieval or clustering stages, rather than the structure module of AF2.

AF-Cluster analysis. Root-mean-squared deviation (RMSD) for structure models was calculated in MDtraj⁸⁶. Principal component analysis and t-SNE⁵⁸ were calculated using Scikit-learn⁸⁷. Spearman correlations and t-tests were calculated using Scipy⁸⁸. Protein structures were visualized in PyMOL⁷⁶.

Data availability. Data corresponding to all AF-Cluster modeling and analysis presented here are publicly available at www.github.com/HWaymentSteele/AF_Cluster. The NMR assignments of KaiB^{RS}, KaiB^{RS}-3m, and KaiB^{TV}-4 have been deposited in the Biological Magnetic Resonance Bank (BMRB) under accession codes 52018, 52017, 31107, respectively. The NMR structure of KaiB^{TV}-4 is available at PDB accession code 8UBH and BMRB accession code 31107.

Code availability. Scripts for running AF-Cluster, AF2, and MSA Transformer, and analysis presented here are available at www.github.com/HWaymentSteele/AF_Cluster.

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Extended Data Figure and Table captions

Extended Data Figure 1: Investigating two highly-similar sets of sequences in the KaiB^{TE} MSA. a)

Sequences 1-50 predict the ground state in all 5 AF2 models, whereas sequences 50-100 predict the FS state in 4 of 5 models. Sequences are ranked by sequence similarity from the ColabFold MSA generation routine. b) MSA Transformer predicted contacts for both sets of sequences. c) Taking the difference of both contact maps highlights that sequences 50-100 contain features for the FS state corresponding to beta-strands (boxed in orange, magenta) and the helix-helix interaction (boxed in red). Right: Structure model (PDB: 5JYT) for the FS state of KaiB^{TE}, features colored analogously.

Extended Data Figure 2: Empirically maximizing information content of clustering using DBSCAN³⁵. a)

Varying the parameter epsilon, which controls the maximum allowable distance for points to be in a cluster, results in a peak in the number of clusters DBSCAN identifies for a set of sequences. b) For epsilon < ϵ^{max} , fewer sequences are clustered, i.e. more are identified as outliers by the DBSCAN algorithm. For epsilon > ϵ^{max} , more sequences are clustered but fewer clusters are returned as more clusters are joined. c) Example clusterings of KaiB sequences at different epsilon values (compare to Figure 1D). d) Corresponding KaiB landscape of predictions for these epsilon values. e) The pLDDT values of models within 3 Å RMSD of the ground and FS state from the clustered sampling method are statistically significantly higher than the rest of the models. Box plots depict median and 25/75% interquartile range, whiskers = 1.5*interquartile range. P-values for sample comparisons with $p < 0.05$ indicated, calculated via a two-sided test for the null hypothesis that 2 independent samples have identical mean values. $n = 500$ models for the two Uniform sampling methods, $n = 230$ for AF-Cluster sampling.

Extended Data Figure 3: AF-Cluster sampling detects KaiB^{TE} ground state in evolutionary couplings. a)

Contacts under 5 Å that correspond uniquely to the FS state (left) or ground state (right). Boxed features correspond to features unique to both states. b) AUC scores to both states of contact maps predicted by MSA transformer, a method trained by unsupervised learning. Randomly-subsampled MSAs have higher score to the FS state, and AF-Cluster contacts have higher score to ground state. c) Contact maps of sampled MSAs with the highest AUC to the ground state from (i) AF-Cluster and (ii) random sampling. The best-scoring random sample does not include the beta-strand unique to the ground state (boxed in blue in i). (iii) Contacts calculated from the whole MSA show features corresponding the FS state: beta-strands (orange, magenta) and the helix-helix interaction (red) boxed in (A). d) Contact map scores for both states correlate to the AF2 prediction RMSD for each state (Ground state: Spearman $R = -0.32$, $p = 2e-09$, FS state: Spearman $R = -0.34$, $p = 4e-10$ via a two-sided statistical test. No adjustment for multiple comparisons was made). Error bands for the linear trendline are 95% confidence intervals obtained via bootstrapping.

Extended Data Figure 4: Supplemental experimental data for KaiB^{TV}-4 and KaiB^{RS}-3m. a)

¹H ¹⁵N HSQC spectra of KaiB^{TV}-4 indicates one major folded state. Assignments are shown. b) Strip plot extracted from a 150 ms mixing time ¹⁵N-edited NOESY-HSQC spectrum of KaiB^{TV}-4 illustrating the inter-strand NOEs between residues V3-V8; T35-D40; T58-Y63; R68-Y71, used in confirming KaiB^{TV}-4 is in the fold-switched state. c) Summary of NOEs between the parallel β -sheets V3-V8 and T35-D40, and the antiparallel β -sheets T58-V62 and R68-V71. Confirmed NOEs are depicted by dashed lines. NOEs not depicted could not be confirmed unambiguously. SEC-MALS analysis of (d) KaiB^{TV}-4 and (e) KaiB^{RS}-3m at NMR concentration of 500 μ M indicate both are monomeric. The profiles on the left show the full SEC-MALS run with the light scattering (LS) profile in blue, normalized UV profile in red, and refractive index (RI) profile in green. On the right is the region of the peak of interest showing the light scattering profile (blue) plotted against elution time, and the protein

molar masses are indicated in red. The molar masses of KaiB^{TV}-4 and KaiB^{RS}-3m have been determined from light scattering and refractometry data to be (9.5 +/- 3.0) kDa and (9.4 +/- 1.7) kDa, respectively.

Extended Data Figure 5: Three mutations are sufficient to switch KaiB^{RS} AF2 prediction to high-confidence FS state prediction. a) pLDDT from AF2 (no MSA, 12 recycles, model 1) for all combinations of 8 possible point mutations most enriched from FS state analysis (cf. Figure 3B). Quadruple-mutants and greater are not labeled by residue mutation, as we searched for the minimal set of mutations to flip the conformational equilibrium. b) Structure models of single mutant V83D, double mutant V83D-I68R, and triple mutant V83D-I68R-N84A demonstrating that V83D switches the C-terminal strand to a helix, and I68R switches the C-terminal helices to a strand. N84A increases the pLDDT of the prediction of the FS state. Top row: structures colored as in Figure 1a. Bottom row: structures colored by pLDDT.

Extended Data Figure 6: Results corresponding to testing AF-Cluster for other proteins. a) Predicting the structure of RfaH in AF2 with the complete MSA from ColabFold³⁴ returns the autoinhibited state with a mean pLDDT of 68.6 (note low confidence in the first alpha-helix of the CTD.) b) B-factors of PDB 5OND⁹⁰, indicating that the last helical turn of the second to last helix has high B-factors (arrow). c) AF-Cluster only predicts the monomeric state for proteins that switch between monomeric and oligomeric states.

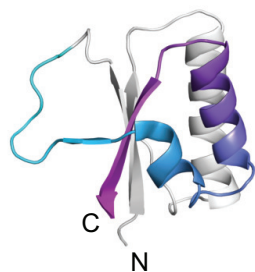
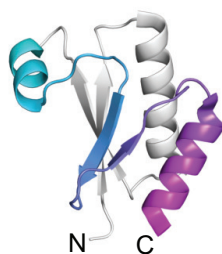
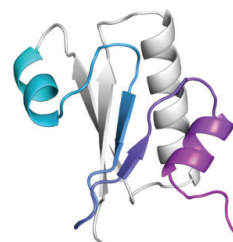
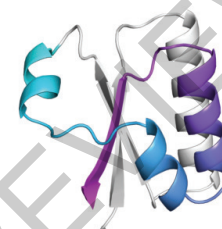
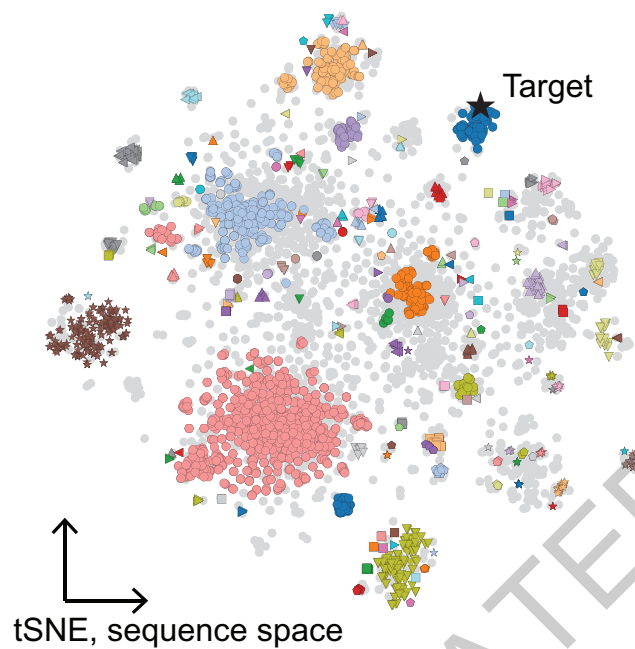
Extended Data Figure 7. Investigating the source of the AF-Cluster prediction for an alternate state of Mpt53. a) pLDDT vs. RMSD for AF-Cluster sampling on oxidoreductase Mpt53. Each prediction colored by MSA size. b) pLDDT values for state 1, corresponding to the known thioredoxin-like state, and an alternate unknown state are significantly higher than background. Box plots depict median and 25/75% interquartile range, whiskers = 1.5*interquartile range. P-values for sample comparisons with $p < 0.05$ indicated, calculated via a two-sided test for the null hypothesis that 2 independent samples have identical mean values. $n = 1642$ models total. c) The conserved CxxC active site is very similar between its conformation in the crystal structure and models for the putative alternate state. d) Workflow for using DALI⁹² to screen for structure homologues to both Mpt53's original state and predicted alternate state to search for any similar structures in the PDB that might have been in AF2's training set. e) Plotting RMSD normalized by alignment length to both structures reveals some structures with lower weighted RMSD to the alternate state than to the original state. f) 7 of 9 DALI hits with lower alternate state RMSD contained an alpha-helix positioned in similar same way as in the Mpt53 alternate state (colored in green). One structure (3EMX) also contained an N-terminus beta-strand positioned similarly to the alternate state.

Extended Data Figure 8: An analogous fold-switch state is predicted for some Mpt53 structure homologues. 6 of the 10 screened homologues from DALI⁹² with the lowest RMSD to the original state predicted an alternate state similar to that of Mpt53. a) Conformational landscapes, visualized by RMSD to two states of Mpt53, and showing the corresponding known structures (above) and predicted alternate structures (below), colored analogously to Mpt53 (cf. Figure 5e). b) Alternate structures in (a), colored by pLDDT. c) Conformational landscapes of 4 structure homologues with no evidence for predicted alternate state.

Extended Data Figure 9: Phylogenetic tree of closest structure matches for Mpt53 states. Homologues for Mpt53 original state and alternate state are dispersed across a calculated phylogenetic tree of the structure hits for both identified via DALI⁹² (cf. Extended Data Figure 7).

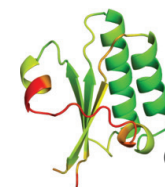
Extended Data Figure 10: MSA clusters enable correct predictions for engineered fold-switching point mutations in the protein G_A/G_B system. A) Sequences of the 12 sets of G_A/G_B point mutations tested, from refs. ⁴⁹⁻⁵¹. Point mutations different from neighboring sequences in the series are colored in orange. Right: Representative NMR structures of the G_A and G_B fold. B) Left: Visualization of sequence identity and coverage of the MSA returned by ColabFold for G_A98. Right: Visualization of MSA clusters with more than 10 sequences from the AF-Cluster clustering routine. C) We compared 3 types of MSAs for each point mutation: i) the full MSA returned by ColabFold, ii) MSA clusters returned by AF-Cluster, and iii) MSAs of the wild-type G_A and G_B proteins in ref. ⁵⁰. Predictions for which the highest pLDDT is incorrect are marked with an X. AF-Cluster has a higher success rate and returns predictions with higher pLDDT.

Extended Data Table 1: Structure data corresponding to the NMR structure of KaiB^{TV}-4 (PDB: 8UBH).

a Ground state**b** Fold-switch (FS) state**c** AlphaFold2,
MMseqs2 MSA: FS stateClosest 50 sequences
in MSA: Ground state**d** Cluster MSA of target sequence by similarity

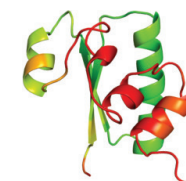
Use clusters as AlphaFold2 input

★ RKTYYVLKLYVAGNTPNSVRALK...
 ● -KTYILRLYVAGTTSRSNKAIT...
 ● -KTYILRLYVAGTTSRSNKAIT...
 ● --TYVLRLYIAGATPQSIKAIT...
 ⋮

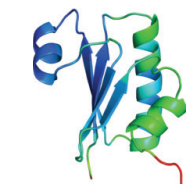


Ground state

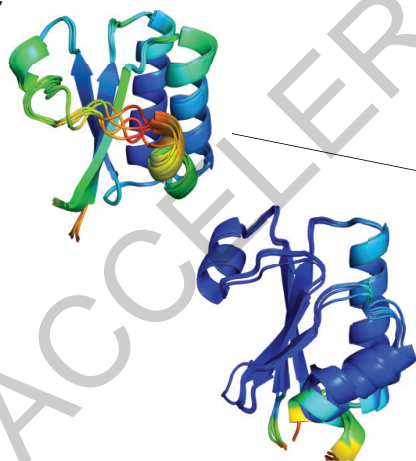
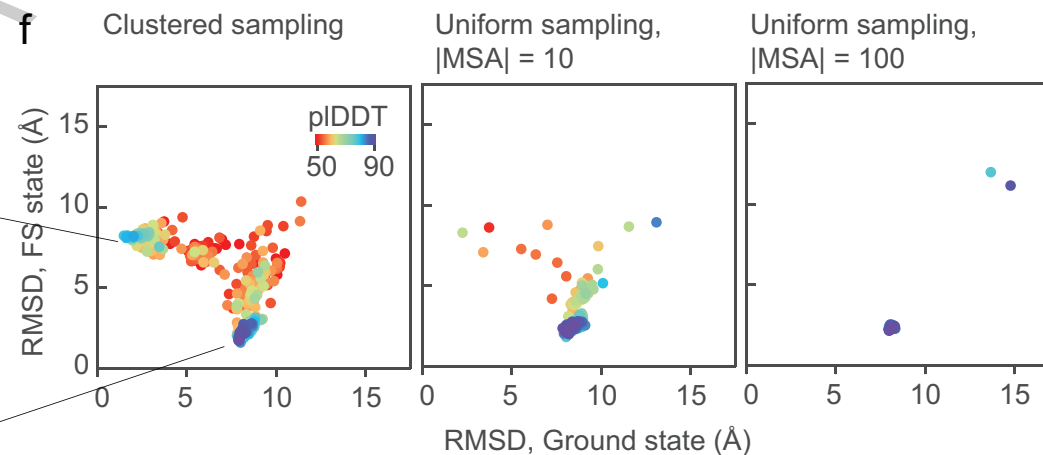
★ RKTYYVLKLYVAGNTPNSVRALK...
 ▲ --YVLRLYVAGMTPRSIEAIS...
 ▲ QQKYVLRLYVAGMTPRSMQAIS...
 ▲ QQYVLRLYVAGMTPRSMQAIS...
 ⋮

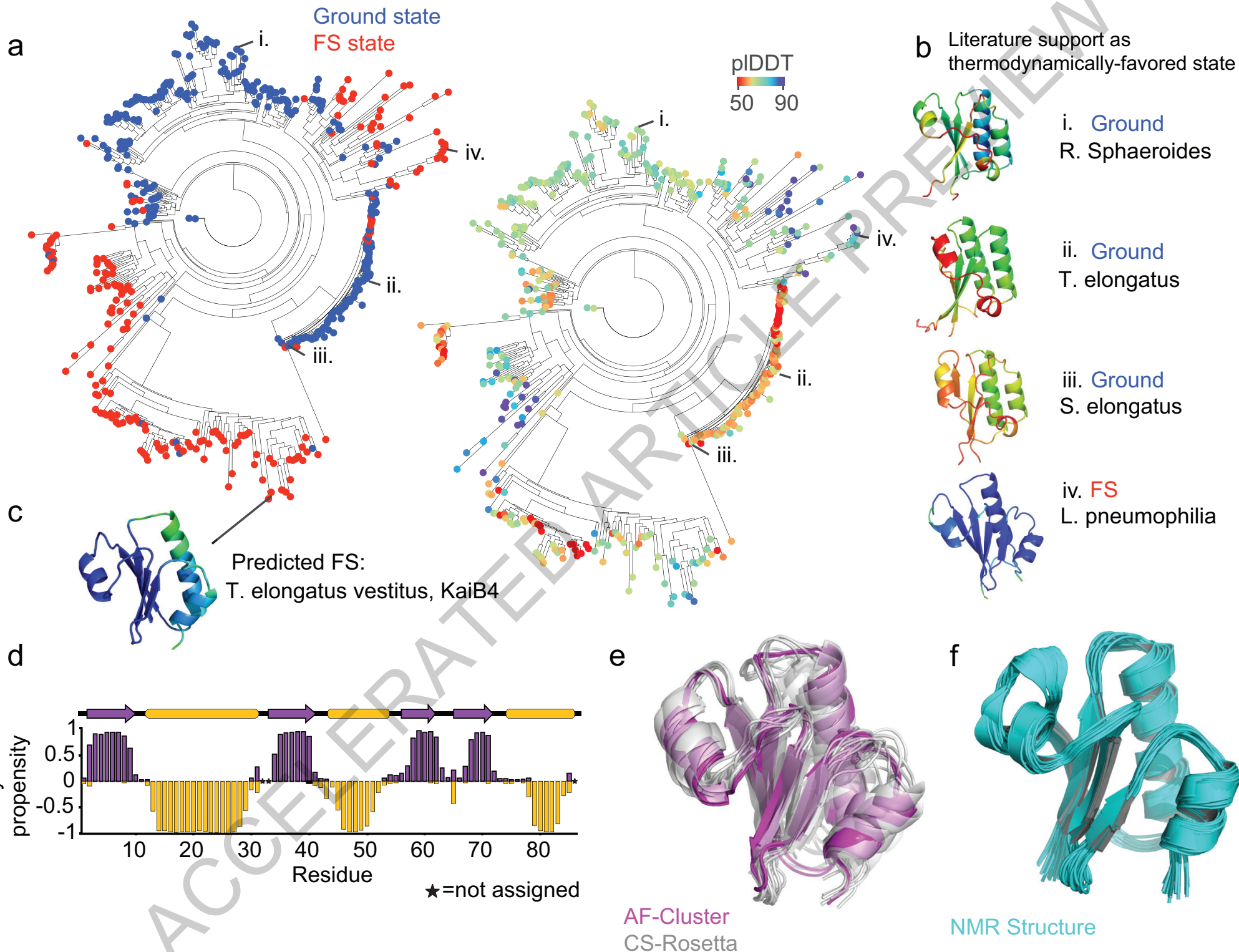


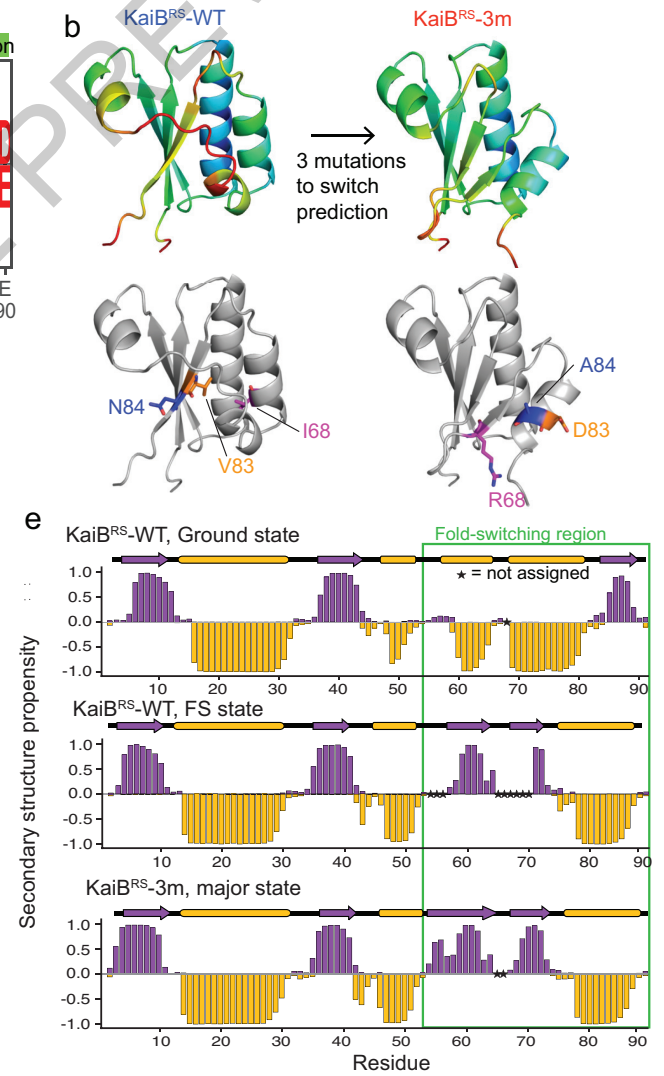
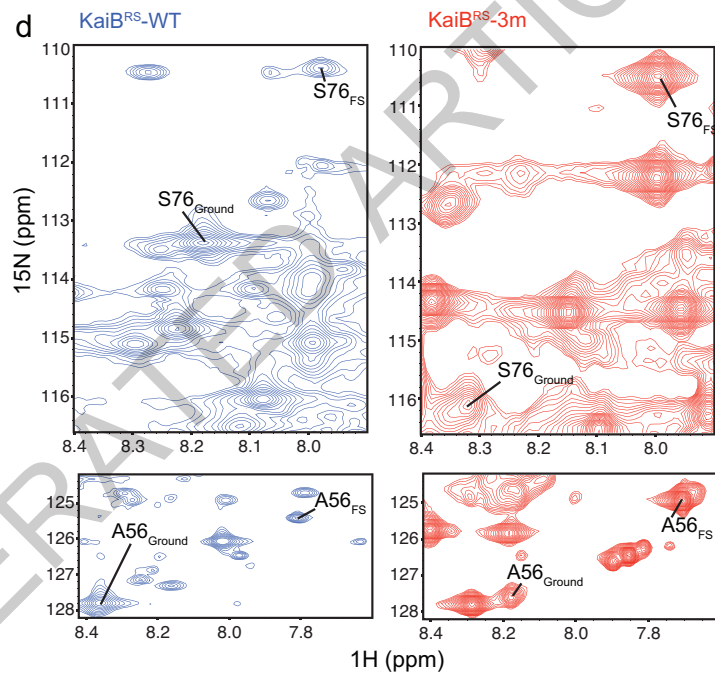
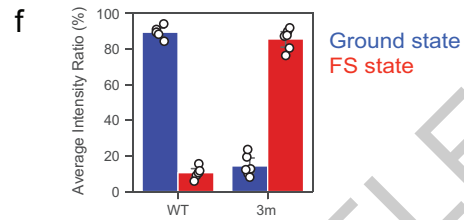
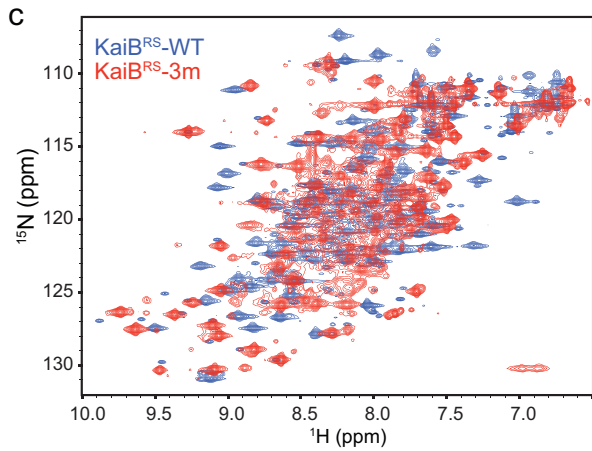
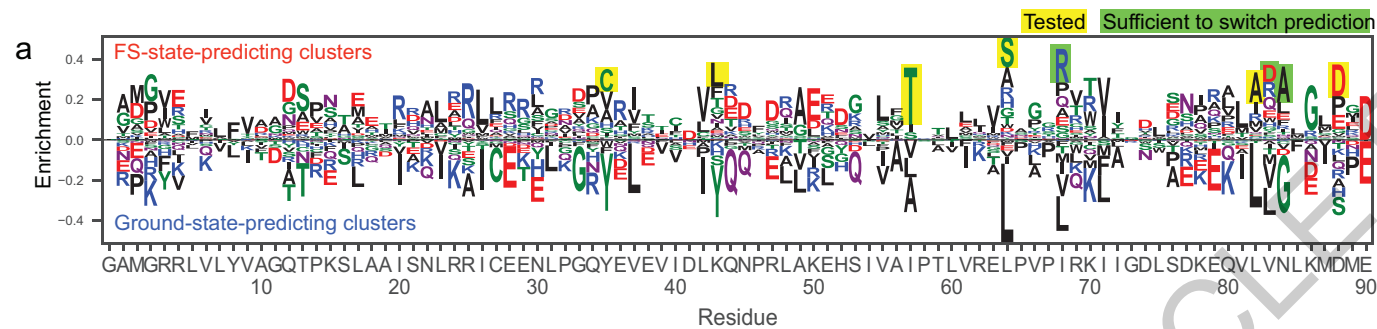
★ RKTYYVLKLYVAGNTPNSVRALK...
 ▼ -PAYVLRLFVAGHSPNTQRILQ...
 ▼ ---YVLRLFVSGYSAATARILQ...
 ▼ ---YILRLFVAGHSPNTQRILQ...
 ⋮



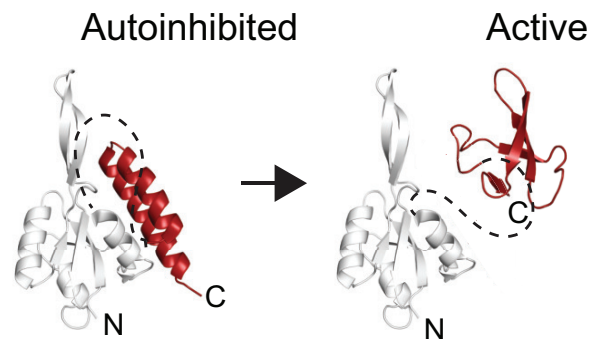
FS state

e**f**

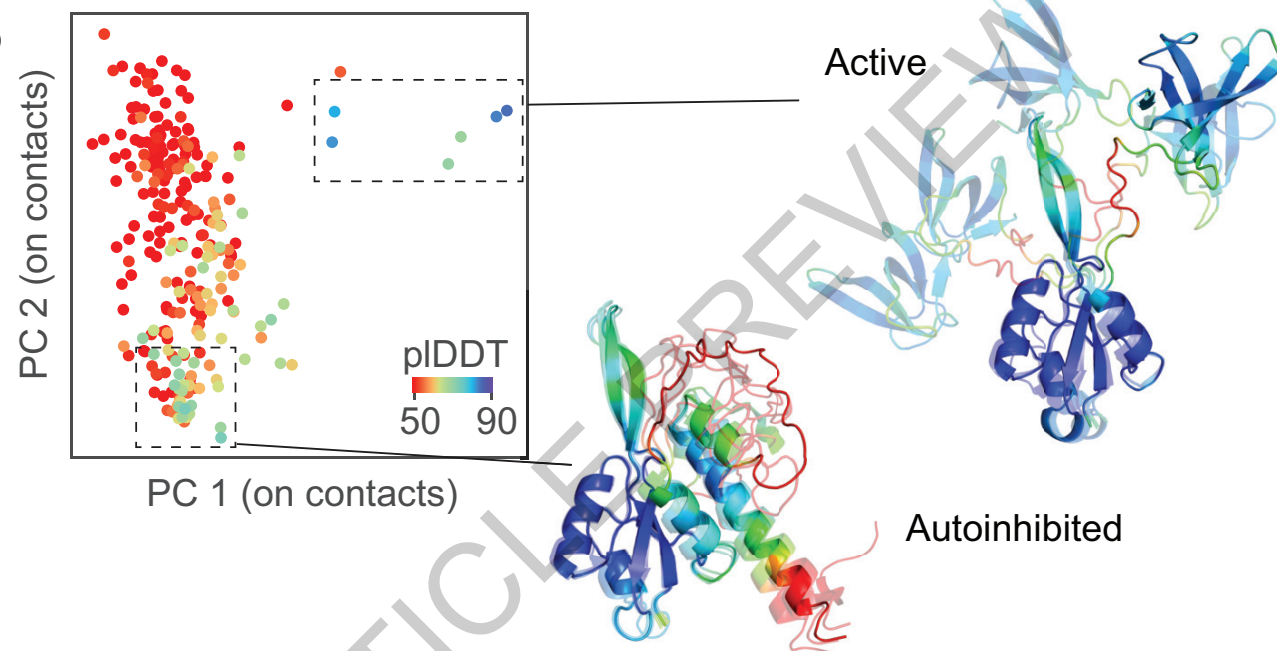




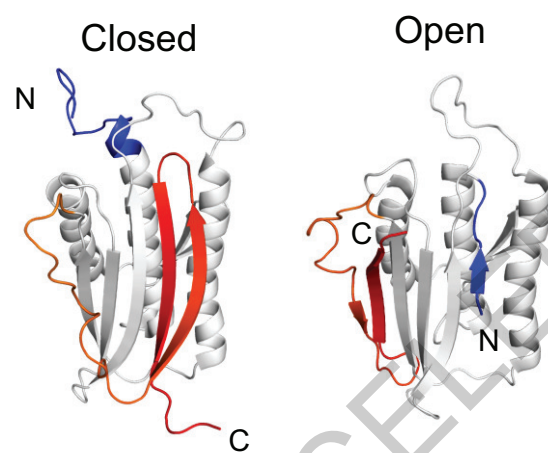
a RfaH transcription factor
E. coli



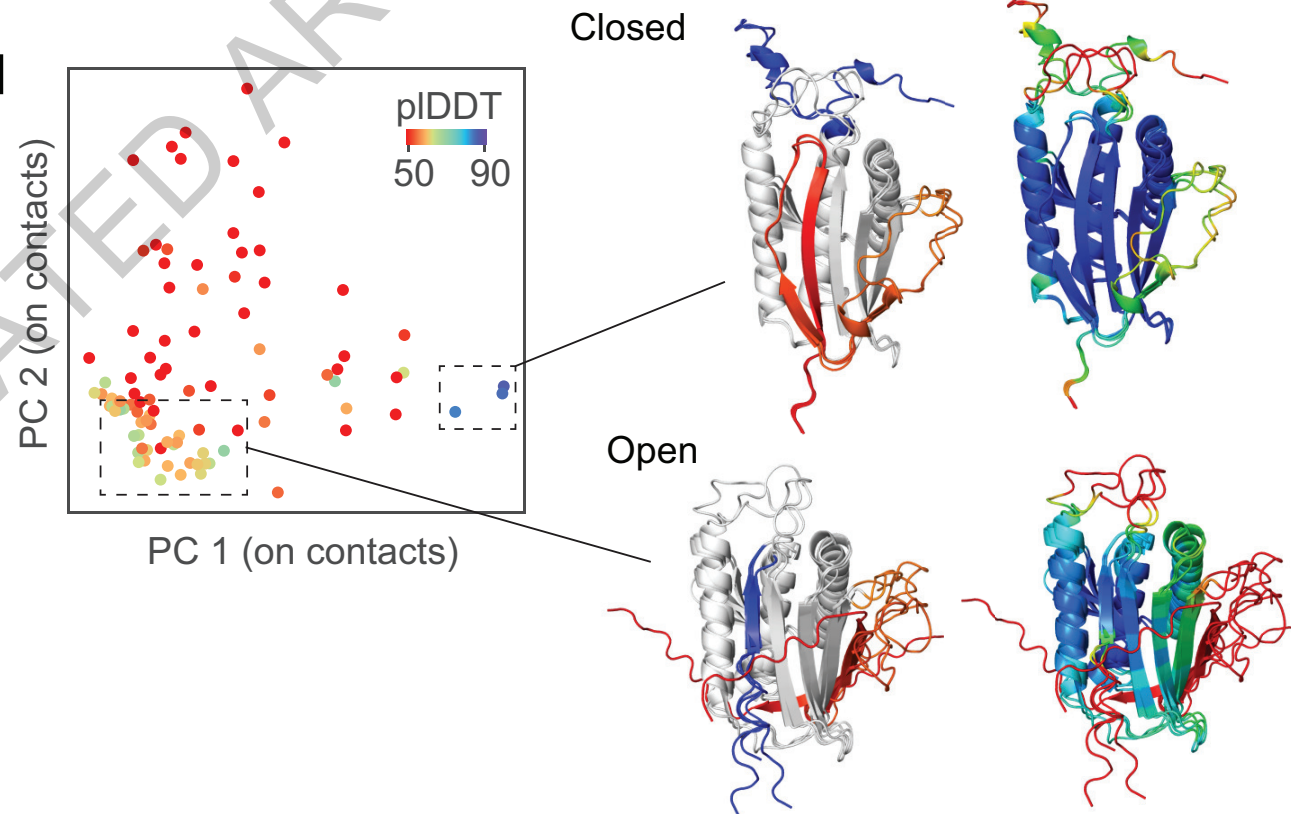
b



c Mad-2 spindle checkpoint
H. sapiens



d



a

AF-Cluster alternate state screening

Initial screen of
628 families,
run AF2 on
10 clusters each

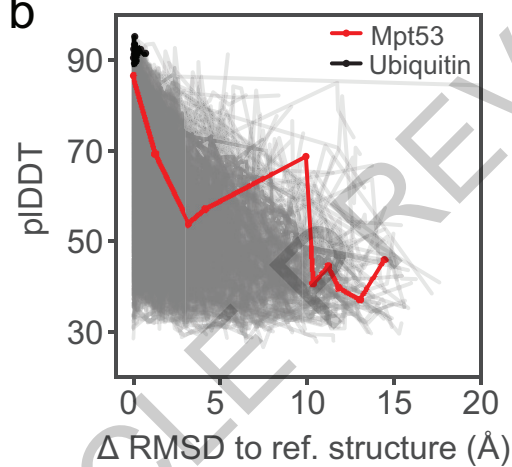


Look for
high pLDDT &
high RMSD

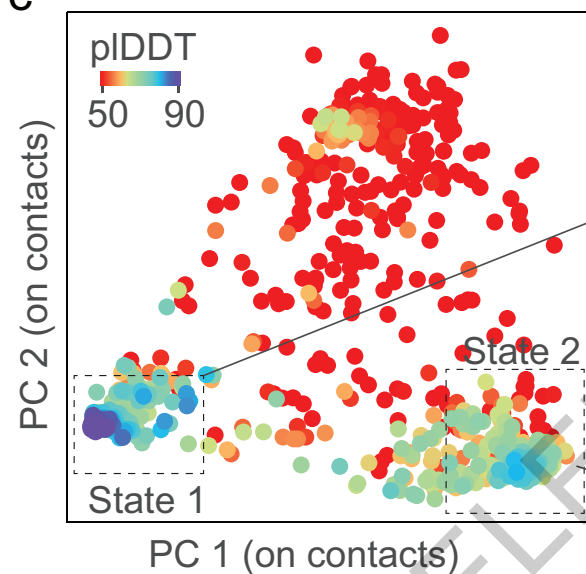


Run AF2
on all clusters
for candidate

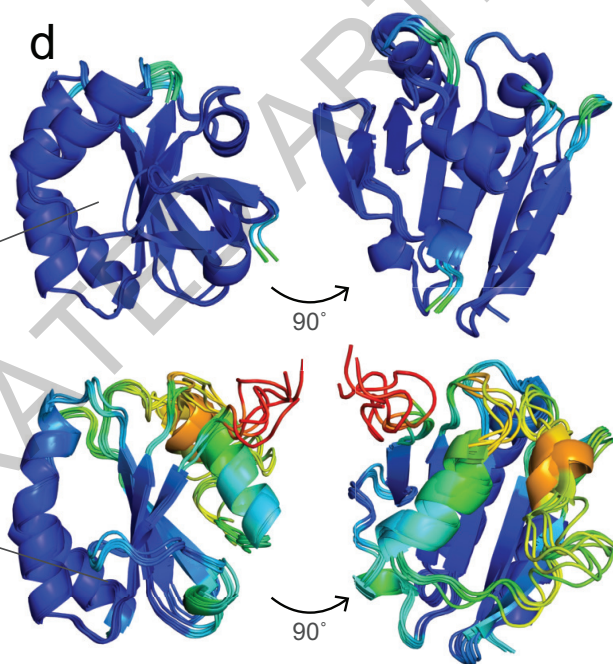
b



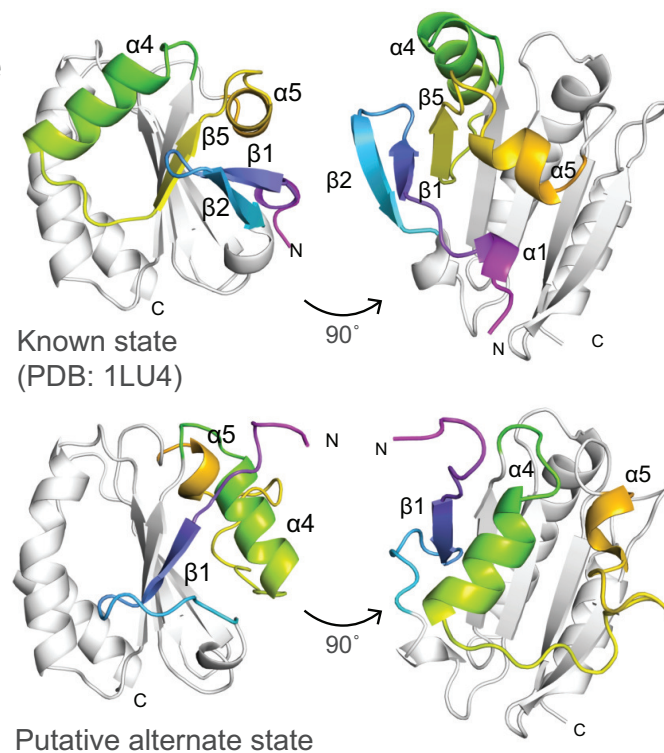
c

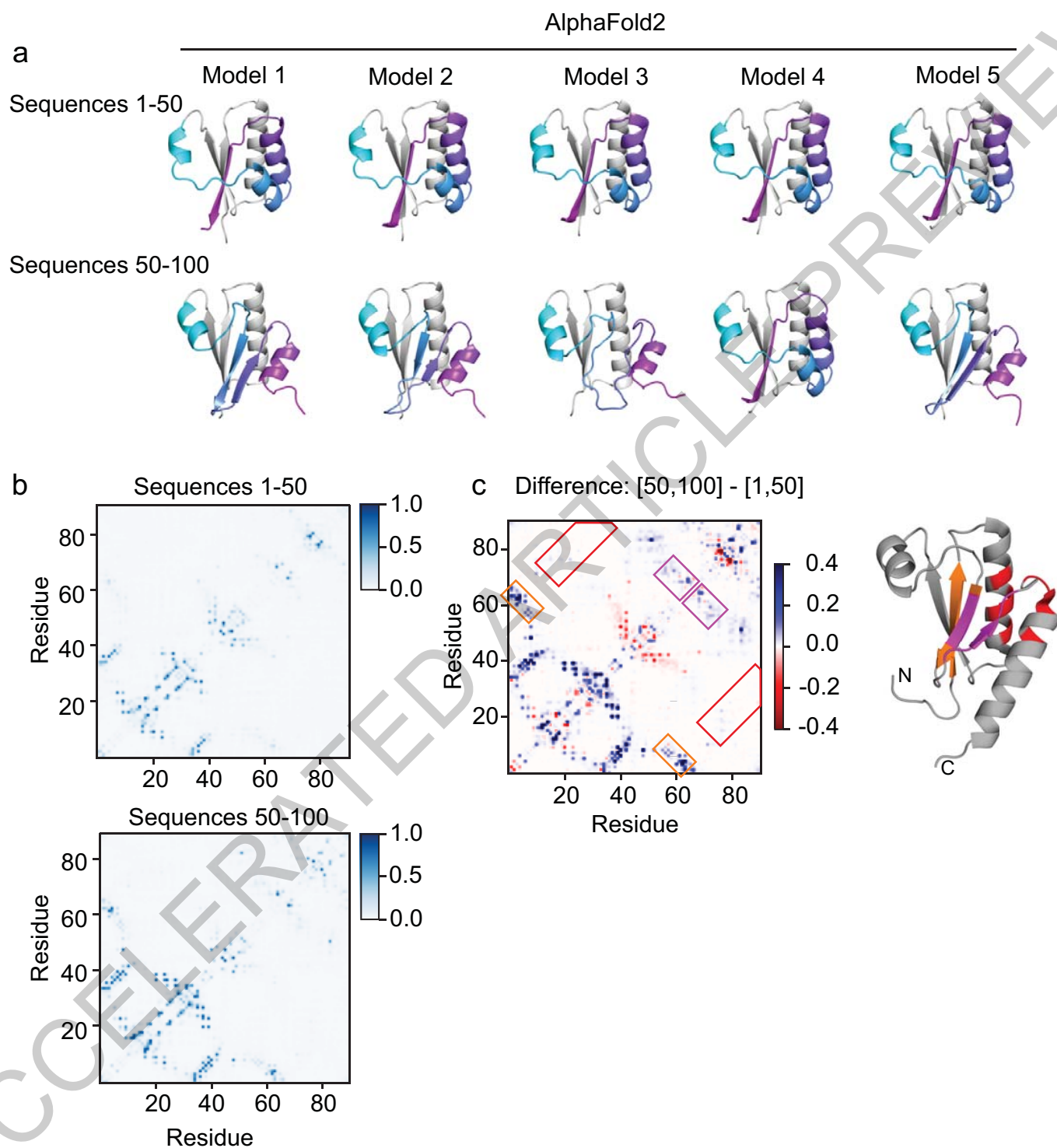


d

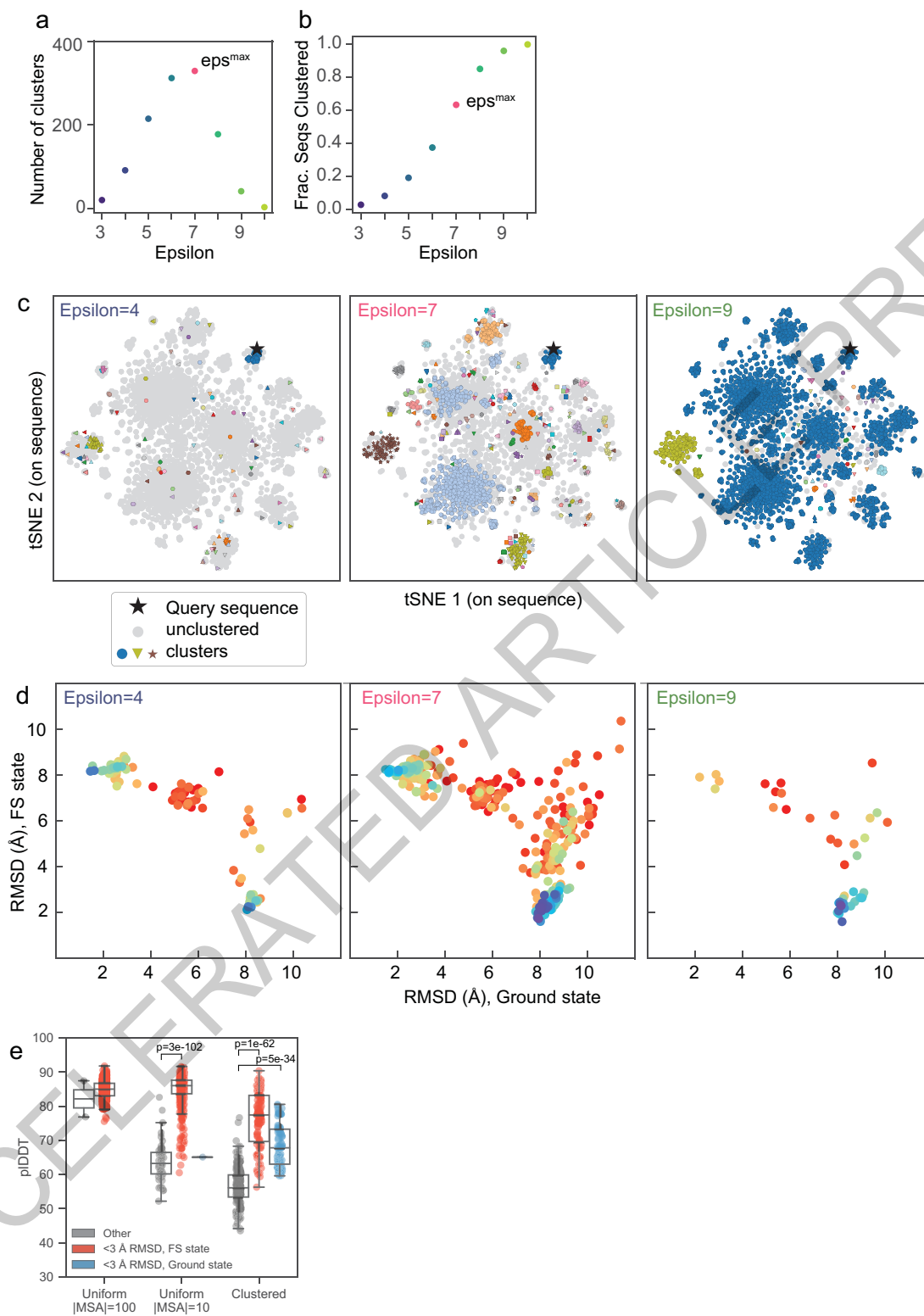


e

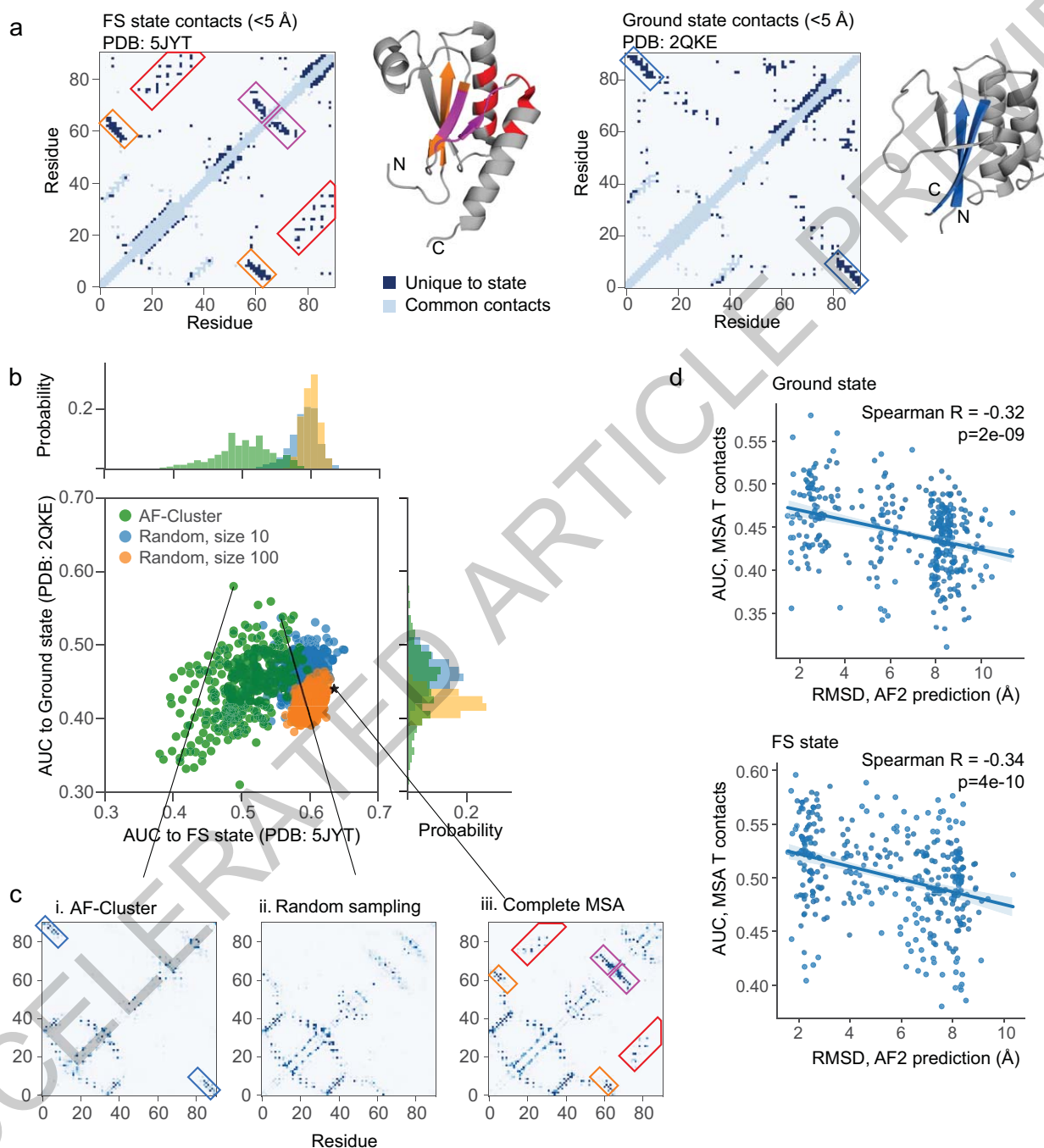




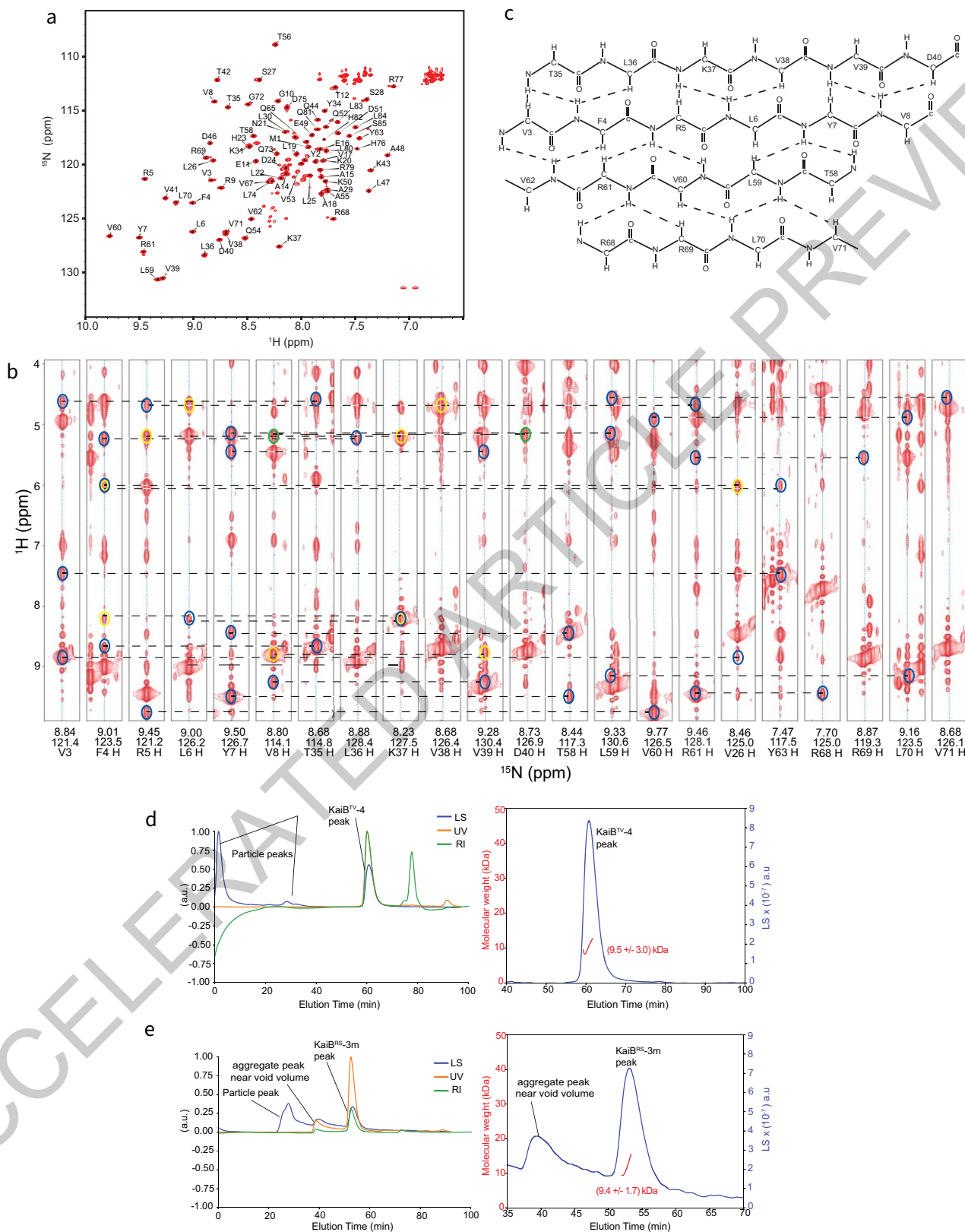
Extended Data Fig. 1



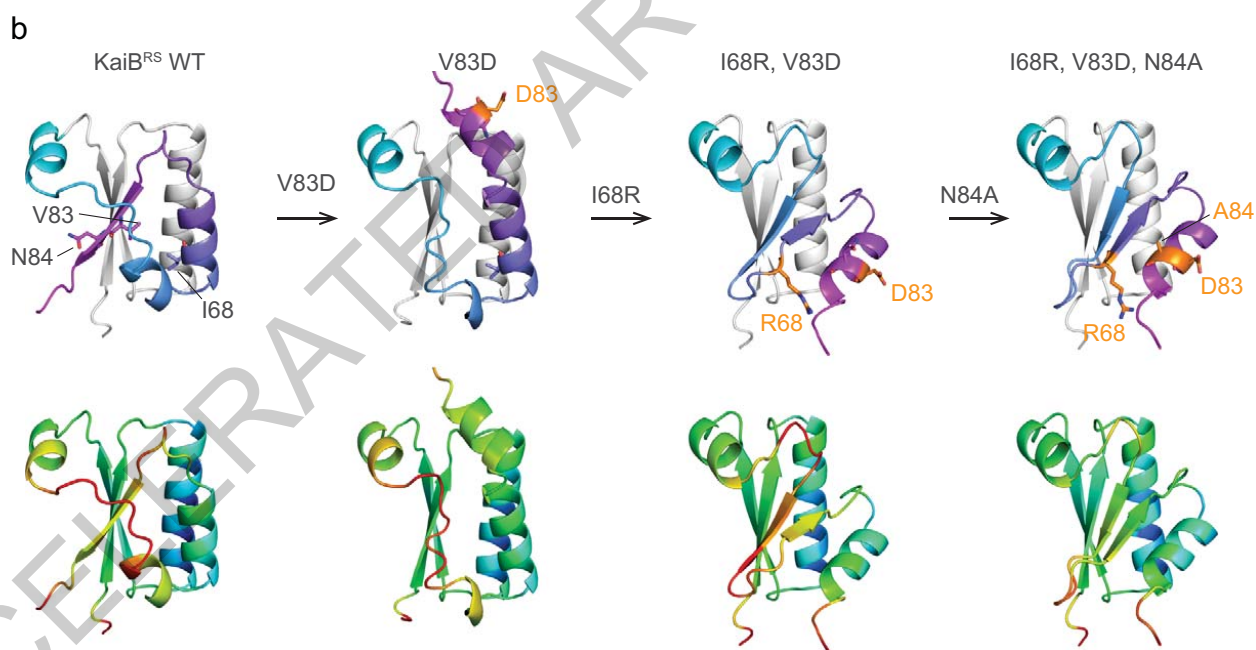
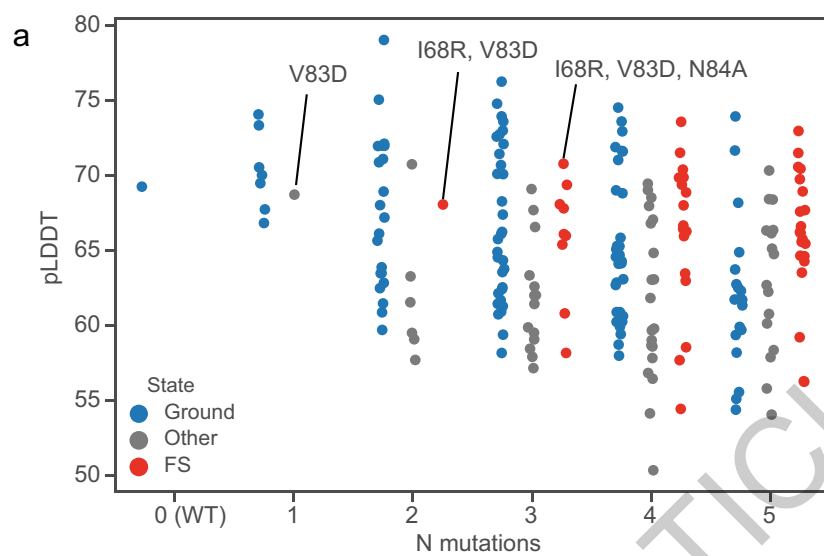
Extended Data Fig. 2



Extended Data Fig. 3



Extended Data Fig. 4

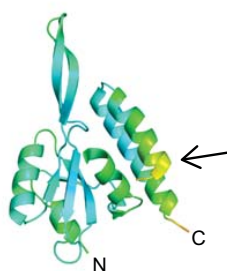


Extended Data Fig. 5

a AF2, complete MSA



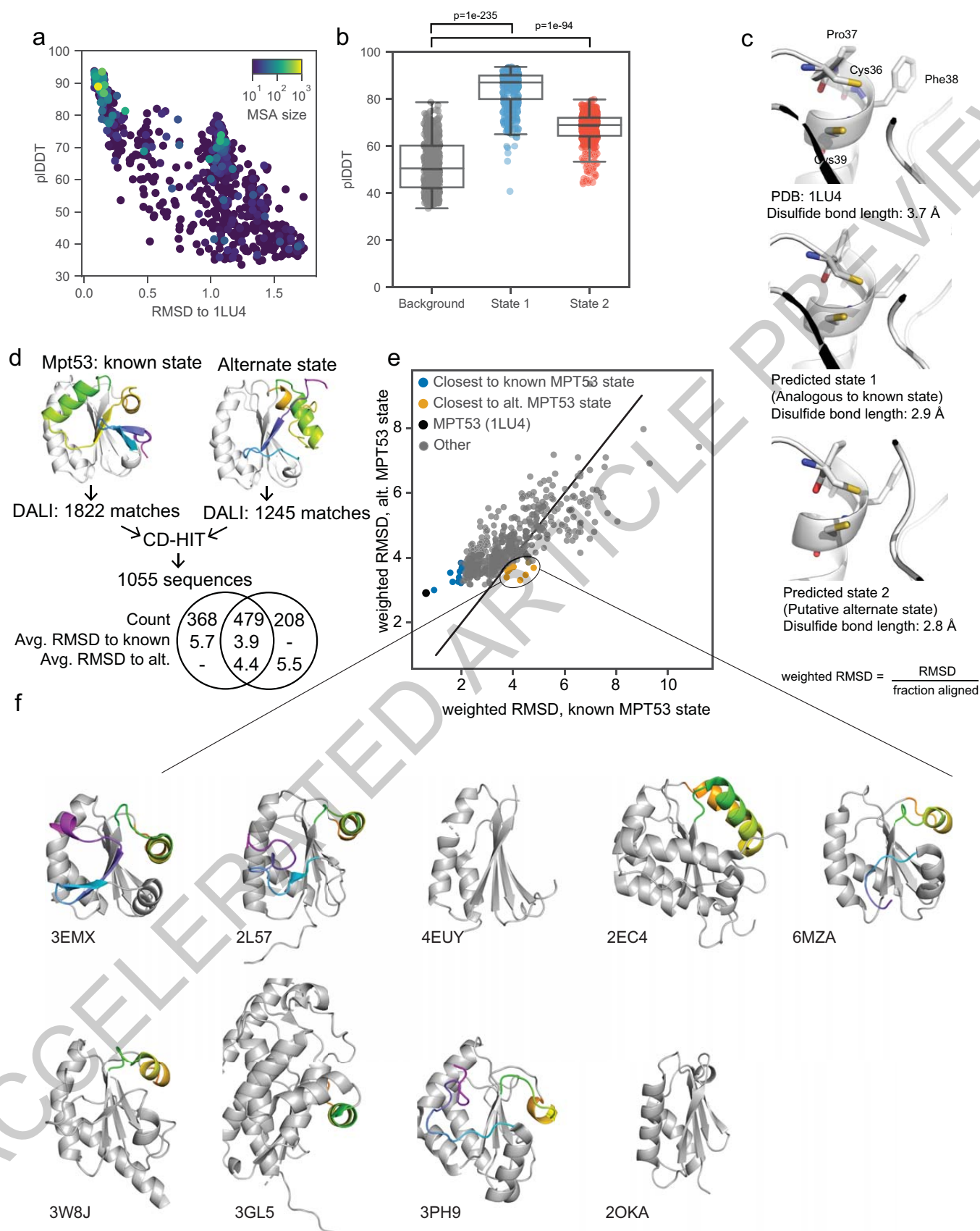
b B-factor



c

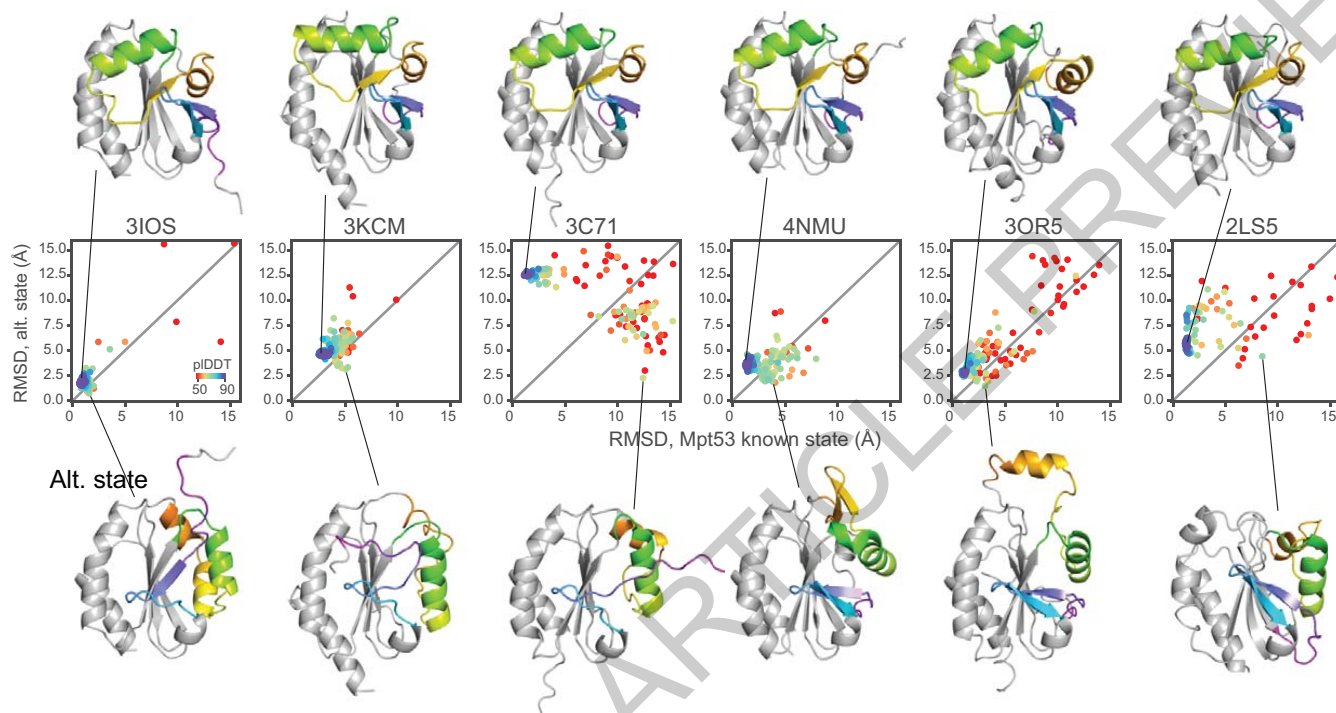
Protein Classification Organism	Monomeric state	Subunit of oligomeric state	Oligomer	AF-cluster models	AF-cluster models, colored by pLDDT
Selecase Metalloproteinase M. janaschii	 4QHF	 4QHH			
Lymphotoxin Cytokine H. sapiens	 1J9O	 2JP1			
CLIC1 Chloride channel H. sapiens	 1K0N	 1RRK			

Extended Data Fig. 6

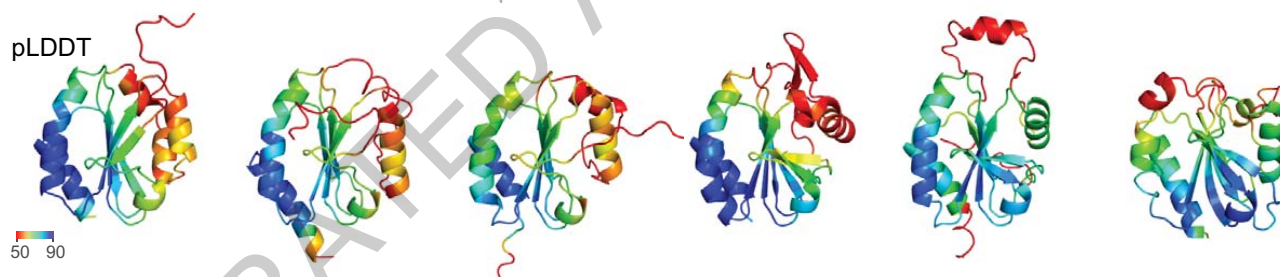


Extended Data Fig. 7

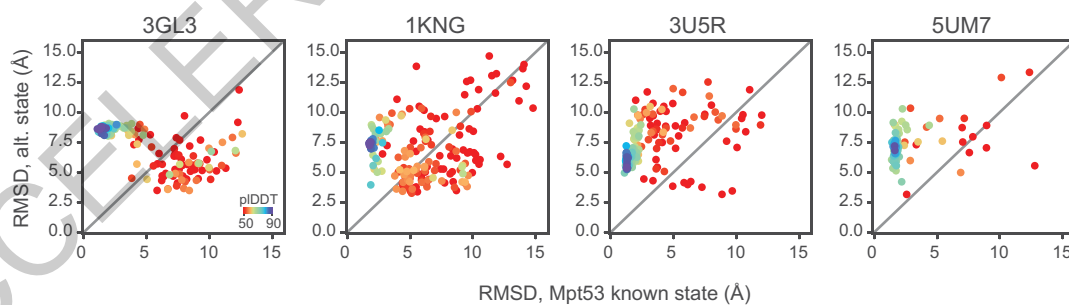
a Known state



b pLDDT

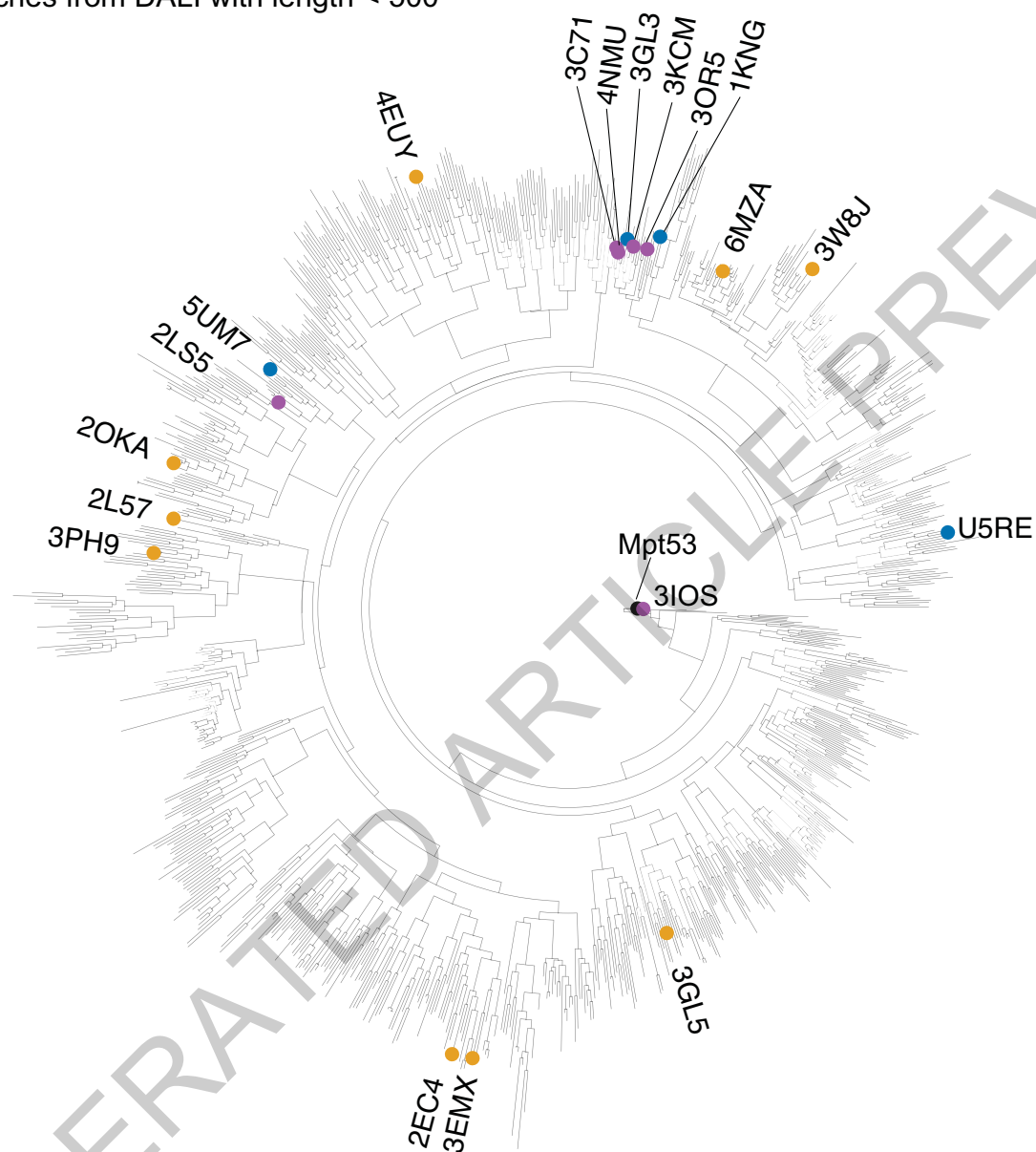


c



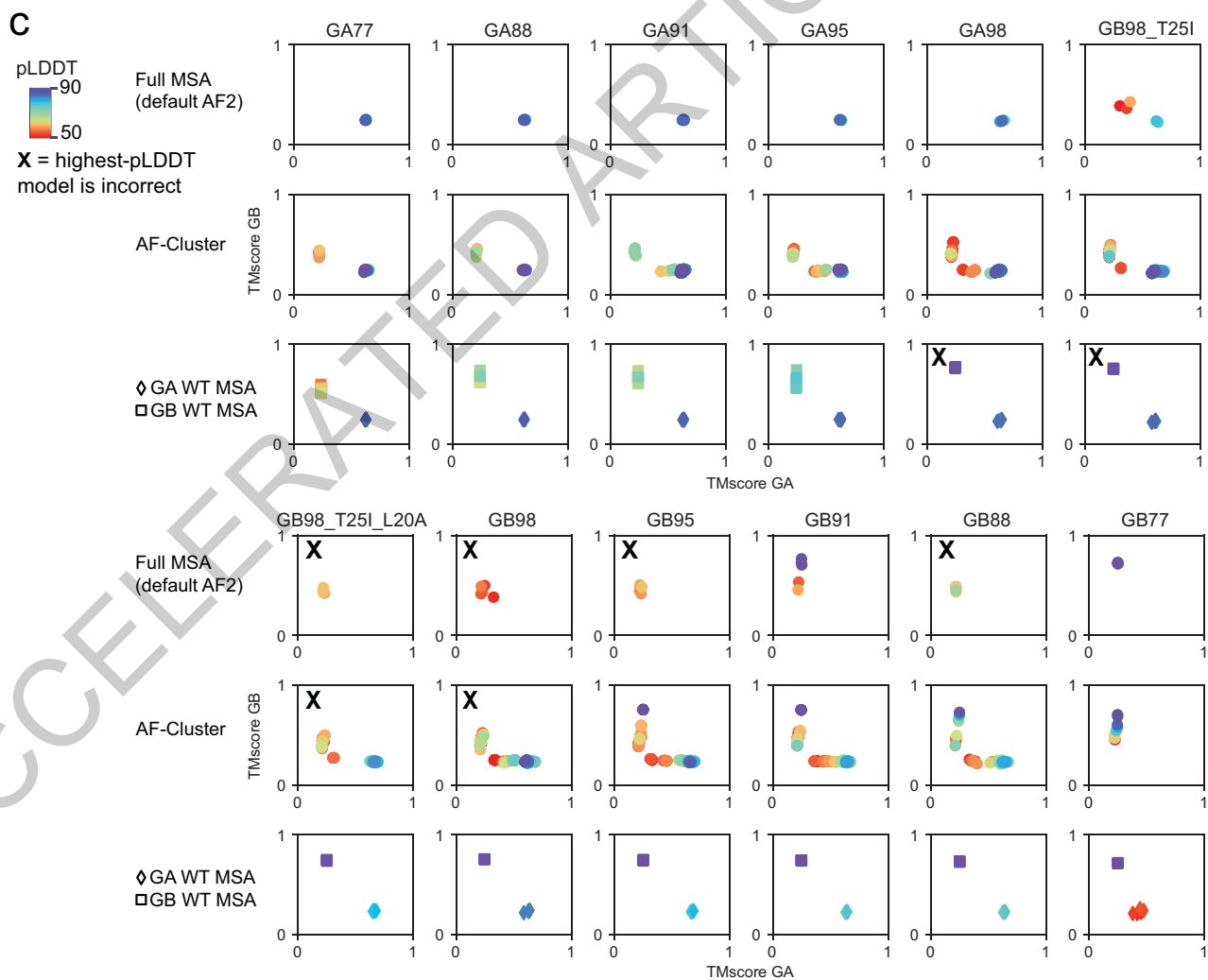
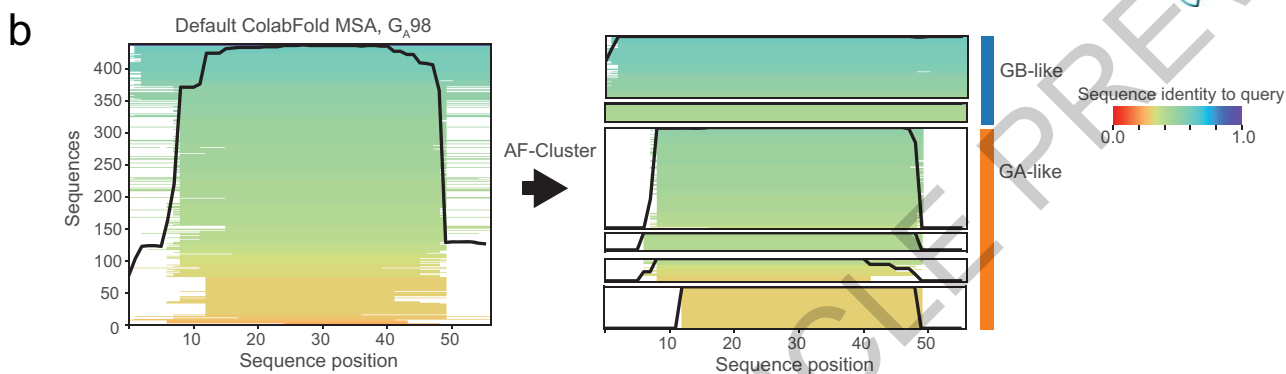
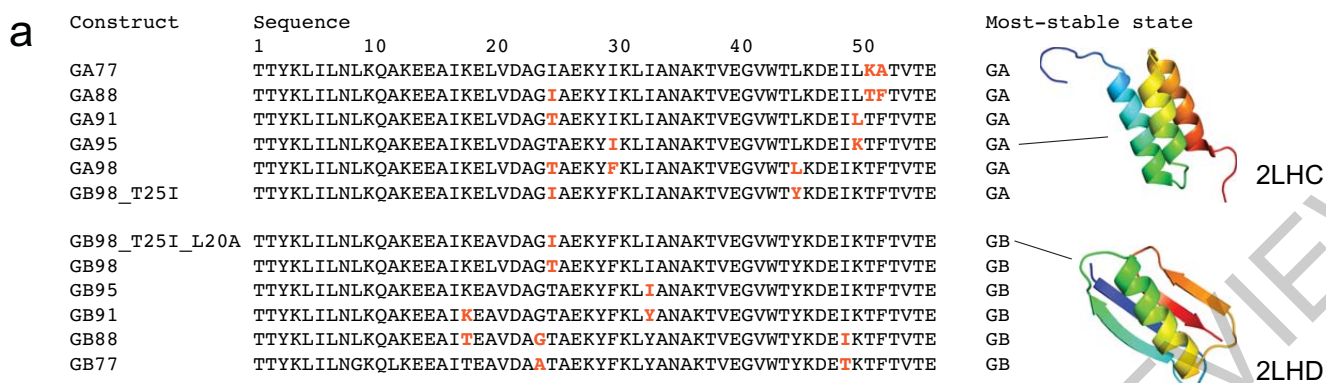
Extended Data Fig. 8

Matches from DALI with length < 500



- Closest to known Mpt53 state, no alt. state in AF-Cluster
- Closest to alt. Mpt53 state
- AF-Cluster predicts alternate state

Extended Data Fig. 9



Extended Data Fig. 10

KaiB ^{TV} -4	
NMR distance and dihedral constraints	
Distance constraints	
Total NOE	622
Intra-residue	163
Inter-residue	
Sequential ($ i - j = 1$)	117
Medium-range ($ i - j < 4$)	75
Long-range ($ i - j > 5$)	205
Intermolecular	N/A
Hydrogen bonds	62
Total dihedral angle restraints	
ϕ	70
ψ	73
Structure statistics	
Violations (mean and s.d.)	
Distance constraints (Å)	0.18 ± 0.07
Dihedral angle constraints (°)	3.02 ± 1.82
Max. dihedral angle violation (°)	8.59
Max. distance constraint violation (Å)	0.46
Deviations from idealized geometry	
Bond lengths (Å)	0.006
Bond angles (°)	0.681 ± 0.013
Impropers (°)	0.410 ± 0.021
Average pairwise r.m.s. deviation** (Å)	
Heavy	1.1
Backbone	0.6

** "Pairwise r.m.s. deviation was calculated among the 20 lowest energy structures."

Average pairwise r.m.s deviations were calculated using secondary structure elements (residues 3-9; 3-28; 35-40; 45-51; 58-61; 68-71; and 76-84)

Extended Data Table 1

Reporting Summary

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For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a Confirmed

- ☐ ☒ The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement
- ☒ ☐ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
- ☐ ☒ The statistical test(s) used AND whether they are one- or two-sided
Only common tests should be described solely by name; describe more complex techniques in the Methods section.
- ☒ ☐ A description of all covariates tested
- ☒ ☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
- ☐ ☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
- ☐ ☒ For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
Give P values as exact values whenever suitable.
- ☒ ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
- ☒ ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
- ☐ ☒ Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection

NMR data collection: VnmrJ version 4.2
Size Exclusion Chromatography coupled to Multi Angle Light Scattering (SEC-MALS) data collection: Astra 8.1.2.1

Data analysis

NMR analysis and structure determination:
NMRPipe: v11.5
SMILE: version 11.5
POKY: build_02-13-2023j
XPLOr: version 3.5
PONDEROSA C/S: Build_04-26-2018
PINE-SPARKY.2: 2018
APES and iPICK: internal to POKY
TALOS-N: Version 4.12
AUDANA internal to PONDEROSA C/S

The AF-Cluster code is publicly available at https://github.com/HWaymentSteele/AF_Cluster. Its dependencies are
scikit-learn v.1.1.1
scipy v1.7.3
Biopython v1.81
mdtraj v1.9.9
numpy v1.21.5
pandas v1.4.3

polyleven v0.8

Phylogenetic tree development:

BLASTP: 2.6.0

MAFFT: v.7

RaxML: v8.2.9

PAML: v.3.3.20170116

CDHIT: v4.8.1

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Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

Data corresponding to all AF-Cluster modeling and analysis presented here are publicly available at www.github.com/HWaymentSteele/AF_Cluster. The NMR assignments of KaiBRS, KaiBRS-3m, and KaiBTV-4 have been deposited in the Biological Magnetic Resonance Bank (BMRB) under accession codes 52018, 52017, 52019, respectively. The NMR structure of KaiBTV-4 is available at PDB accession code 8UBH and BMRB accession code 31107.

Research involving human participants, their data, or biological material

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N/A

Reporting on race, ethnicity, or other socially relevant groupings

N/A

Population characteristics

N/A

Recruitment

N/A

Ethics oversight

N/A

Note that full information on the approval of the study protocol must also be provided in the manuscript.

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Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

No statistical methods were used to determine sample size.

Data exclusions

Residues with significant overlap in the HSQC spectra were excluded from population estimates for the KaiBRS and KaiBRS-3m constructs in figure 3e. This data exclusion practice was pre-established.

Replication

Beyond the first KaiB example, the AF-cluster was tested on a diverse sample of 5 other experimentally-validated fold-switching proteins. NMR experiments were performed once.

Randomization

No randomized sample selection was applied.

Blinding

Blinding was not relevant for this study as the work was done with protein samples of known compositions.

Reporting for specific materials, systems and methods

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Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging